

Article

OAFBC-Net: A Lightweight RGB-Depth Fusion Network for Occlusion-Aware Apple Detection in Complex Orchard Canopies

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Abstract

Accurate perception of occluded apples in tree canopies is important for precision orchard management and robotic harvesting, but RGB-only vision is sensitive to leaf occlusion, fruit overlap, and unstable illumination. This study presents OAFBC-Net, a lightweight RGB-depth network for joint localization and occlusion-level recognition of apples classified as non-occluded (NO), soft-occluded (SO), or hard-occluded (HO). OAFBC-Net integrates three components: a Fourier amplitude-phase enhancement fusion module (FAPEF) for cross-modal frequency-domain fusion, a weighted bidirectional feature pyramid (BiFPN) for multi-scale interaction, and a Triple Context Guided Block (TCG) for local-to-global semantic compensation. Using an orchard RGB-depth dataset with 3000 image pairs and 134,999 annotated instances acquired with an Intel RealSense D455 camera, OAFBC-Net achieves AP50 values of 72.1%, 87.3%, and 57.9% for SO, NO, and HO, respectively. Under the same split, it outperforms the self-implemented RGB-depth baselines. With FAPEF, BiFPN, and TCG enabled, the macro-average mAP50 and mAP50–95 reach 72.4% and 55.2%, respectively. The model has 3.6 M parameters. On a single image, the total latency is 11.3 ms, corresponding to an inference speed of 88.5 FPS. Three random seeds and five-fold cross-validation with a fixed independent test set show consistent gains over the dual-modal baseline. These findings support OAFBC-Net as a promising visual perception method for occluded-fruit detection in orchards, while cross-site and field-robot validation remain necessary for deployment-oriented claims.

Keywords: occluded apple detection; RGB-depth fusion; orchard canopy; frequency-domain fusion; context modeling; precision horticulture; robotic harvesting



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1. Introduction

Apples play an important role in the global fresh-fruit supply chain and orchard economy [1]. As large-scale cultivation and intensive management continue to advance, higher requirements are imposed on fruit quality, yield, and harvesting efficiency [2]. Field operations still rely heavily on manual labor [3], while rising labor costs and structural labor shortages constrain the long-term competitiveness and sustainable operation of the apple industry [4]. At the same time, the increasing maturity of sensing technology, machine vision, machine learning, and robot control in complex environments greatly accelerates the transition of orchard robots and intelligent operation systems from experimental prototypes to field validation and demonstration applications [5,6].

In applications such as mechanized harvesting, grading equipment, and picking robots, stable and accurate perception of apples on trees is a key prerequisite for path planning and operational decision-making [7]. Canopy occlusion and fruit overlap restrict the visible regions of fruits and lead to incomplete boundaries, which reduces localization reliability. These factors further affect the approach trajectory planning, grasping posture selection, and obstacle-avoidance safety of end effectors, and they also weaken the consistency of fruit counting and yield estimation [8]. However, in real orchard scenes, drastic illumination changes and overlapping branches and leaves make the background highly complex. In addition, the similarity between fruits and leaves in color and texture makes robust detection and stable localization under occlusion still highly challenging [9].

Object detection simultaneously performs localization and category discrimination, and it is a fundamental step in orchard fruit perception and robotic operation. Existing methods can be divided into two-stage frameworks, which first generate candidate regions and then perform classification and regression, and one-stage frameworks, which directly conduct end-to-end regression. The latter offers higher inference efficiency and easier deployment, with the YOLO series being a representative example [10,11]. Many studies on agricultural fruit detection build datasets and models based on visible-light RGB images, and they continuously optimize one-stage detection frameworks for complex orchard backgrounds, scale variations, and edge deployment constraints. Qiu et al. [12] use YOLOv11n as the baseline for orchard maturity recognition. While maintaining efficient end-to-end inference, they introduce BiFPN to strengthen multi-scale feature fusion, adopt GhostConv and pruning to compress the model size, and combine an improved IoU-based loss to enhance regression accuracy, thereby achieving lightweight deployment for apple localization and binary maturity classification. Based on YOLOv8, Z. Liu et al. [13] propose Faster-YOLO-AP for single-class apple detection. They address redundancy by constructing a smaller-scale backbone and designing an efficient convolution module, PDWConv, together with a replacement regression loss. Their method achieves a favorable accuracy-efficiency balance in canopy scenes with low parameter and computational costs, confirming the adaptability of lightweight one-stage models to orchard edge devices. To alleviate missed detections caused by occlusion and overlap, Fu et al. [14] introduce a new feature scale and a structural enhancement module into the YOLOv10 framework. By combining attention mechanisms and localization loss improvement, they significantly improve the detection performance for occluded apples while maintaining real-time inference on embedded platforms. H. Yang et al. [15] focus on the difficulties of occlusion, overlap, and small-object detection in high-density orchards. They systematically improve YOLOv7 from data pre-processing, lightweight backbone design, and neck fusion structure to the detection head and loss function, which significantly improves recognition performance under complex illumination and occlusion conditions. Wang and He [16] address young fruit detection before thinning by introducing channel pruning and transfer learning into YOLOv5s. Their method greatly reduces model size and parameter count while maintaining high accuracy, confirming the potential of RGB-based one-stage models for fast deployment on portable terminals. Ma et al. [17] study apple detection across multiple growth stages and combine YOLOv8n with ShuffleNetv2, Ghost modules, SE attention, and an improved regression loss. Their method jointly improves accuracy, speed, and lightweight performance, and it shows good real-time capability on edge devices. Han et al. [18] further integrate CNN and GCN within the YOLOv8n framework. By jointly modeling local textures and global relationships, they enhance robustness in severe weather and occlusion-overlap scenes while preserving inference efficiency. Overall, existing studies show that RGB-based one-stage detection frameworks represented by YOLO have high practical value for apple target recognition. They also form a clear optimization path, namely improving detec-

tion accuracy and edge deployment efficiency in complex orchard environments through multi-scale feature enhancement, lightweight structural reconstruction, attention or relation modeling, and regression loss improvement. Nevertheless, in real scenes where canopy occlusion, fruit overlap, and strong illumination fluctuations coexist, RGB perception still mainly relies on appearance information. It is easily disturbed by shadows, highlights, and fruit-leaf similarity, which leads to missed detections, false detections, and localization jitter. Its limited ability to represent depth order and three-dimensional geometric relationships also further affects the stability of end-effector approach, grasping, and obstacle-avoidance decisions in picking robots [19]. Therefore, relying only on RGB is insufficient to continuously satisfy the need for highly robust perception under complex canopy conditions, and it is necessary to introduce multimodal schemes that can supplement spatial geometric information [20].

To improve perception robustness in complex orchard environments, agricultural vision research gradually shifts from single-sensor schemes to multi-sensor collaboration. Existing studies show that multimodal information fusion can improve image representation quality, feature extraction capability, and detection accuracy to some extent. RGB-depth can provide geometric structure cues by capturing distance and surface undulation information. Tang et al. [21] construct a recognition-localization framework based on RGB-D and improved SOLOv2. They use RGB images to extract semantic appearance features, use depth to constrain spatial structure, and combine camera intrinsic parameters to solve three-dimensional coordinates. Their method outperforms the original baseline in both segmentation and localization metrics. Liu and Hao [22] propose YOLOfuse based on YOLOv5s. Through dual-branch feature extraction, weighted fusion, coordinate attention, and Soft-NMS, their method improves detection performance in dense overlapping scenes and outperforms single-RGB input while maintaining real-time speed. Beyond RGB-D, RGB-T enhances the separability between targets and backgrounds through temperature differences. Cao et al. [23] combine RGB, depth, and thermal imaging information in mango detection, improving target separation under illumination fluctuation and occlusion. Lin et al. [24] adopt cold-stimulation thermal imaging for apple bruise detection, which creates an observable temperature difference between bruised regions and healthy tissues and significantly improves recall and detection accuracy. RGB-NIR and spectral fusion supplement visible-light representation through narrow-band reflectance information. Ünal et al. [25] find that NIR provides more stable representation of subcutaneous damage under low-light and low-contrast conditions. Luo et al. [26] fuse RGB and NIR information for citrus defect detection and achieve higher detection accuracy and recall in complex backgrounds, with more evident improvement for small targets. In general, different modalities have their own advantages and constraints in acquisition cost, environmental sensitivity, and information complementarity. Thermal infrared and near-infrared or spectral modalities are valuable in low-illumination or low-contrast tasks, but their performance is strongly affected by imaging conditions and task scenarios. In contrast, the depth modality directly provides spatial structure and relative distance information, making it more suitable for alleviating geometric ambiguity under canopy occlusion, fruit overlap, and appearance similarity.

Multimodal fusion is commonly divided into early, mid-level, and late fusion [27,28]. Early fusion completes cross-modal alignment and concatenation at the input layer, which helps preserve fine-grained appearance and geometric details. However, it is more sensitive to registration errors and sensor noise, and it may increase computational and bandwidth costs [29]. Mid-level fusion conducts cross-modal feature interaction in the backbone or neck network and is more likely to form joint representations for multimodal recognition tasks [30]. It is therefore widely adopted in multimodal learning tasks, but its actual

effect depends heavily on feature alignment and the design of interaction modules [31]. Wang et al. [32] systematically compare pixel-level, feature-level, and decision-level strategies for visible-infrared object recognition. They point out that feature-level fusion can retain classification-related discriminative information while controlling computational cost. They further use deep mutual learning to enable cross-modal feature interaction and combine it with weighted decision-level fusion to produce final outputs. Their experiments show that this strategy improves recognition performance under visible-light and infrared conditions. For honey origin traceability and component prediction, Z. Liu et al. [33] compare data-level and feature-level fusion. Their results show that, based on spectral preprocessing and task-oriented feature extraction such as partial least squares, feature-level fusion achieves better classification and regression performance. These studies jointly indicate that the benefits of mid-level fusion depend not only on the fusion position, but also on cross-modal alignment, task-oriented feature extraction and interaction mechanisms, and the matching between subsequent classifiers or fusion strategies. If alignment is inconsistent or the interaction design is insufficient to describe the complementarity of heterogeneous modalities, the fusion advantage becomes clearly limited. Late fusion integrates the outputs after independent inference from different modal branches. It is flexible in engineering implementation and has a low cost for module replacement, but it is difficult to exploit fine-grained complementary information from intermediate layers. Yang et al. [34] adopt a CNN-LSTM-MLP structure for harvest-stage prediction of greenhouse melons. They separately model the RGB image branch, which is based on YOLOv8n detection and morphological estimation, and the environmental or irrigation time-series branch, and then complete fused prediction through an MLP after high-level feature concatenation. The CPROS method proposed by Li et al. [35] trains and infers visible-light and infrared branches separately, and fuses class probabilities and detection boxes at the output layer. This design reduces dependence on strict pixel alignment, feature alignment, and large-scale paired data, and it facilitates combination across different detection backbones. The authors also note that, in some small-sample cases or scenarios with high requirements for fine-grained localization, its performance may be inferior to methods that rely on intermediate feature interaction. In summary, for orchard scenes with severe occlusion and illumination changes, the fusion strategy must balance accuracy, robustness, and real-time performance to satisfy the requirement of robotic operations for stable perception [36].

Building on the above discussion, this study targets robust joint localization and harvesting-oriented occlusion-level recognition of apples (NO, SO, and HO) in complex orchard canopies through mid-level RGB-depth fusion.

The aim of this research is to develop a lightweight RGB-depth detector, OAFBC-Net, that improves occlusion-aware apple perception for orchard monitoring and robotic harvesting while maintaining practical real-time inference.

To achieve this aim, the specific objectives are as follows: (1) To develop a lightweight RGB-depth detection framework for joint apple localization and harvesting-oriented occlusion-level classification in natural orchard scenes. (2) To investigate mid-level cross-modal fusion and context-aware multi-scale representation strategies that improve perception of partially occluded apples. (3) To validate the proposed approach on an orchard RGB-depth dataset through comparisons with unimodal and dual-modal baselines, ablation and cross-validation analyses, and supplementary assessments of runtime efficiency and depth-sensor effects under varying illumination.

Based on these objectives, the main contributions are as follows: (1) OAFBC-Net, an integrated RGB-depth detection framework for joint localization and NO/SO/HO classification in complex orchard canopies. (2) FAPEF, a Fourier amplitude-phase enhancement

fusion module deployed at P4/P5 for adaptive frequency-domain fusion of RGB and depth features under occlusion. (3) A context- and multi-scale enhancement design combining BiFPN with TCG, together with extensive experiments showing consistent improvements over a dual-modal concatenation baseline.

The remainder of this article is organized as follows: Section 2 describes OAFBC-Net and the dataset; Section 3 reports the experimental results; Section 4 discusses the findings and limitations; and Section 5 concludes the study.

2. Materials and Methods

2.1. Overall Framework

To improve occlusion-aware apple detection in complex orchard scenes, this study proposes OAFBC-Net (Figure 1). The network detects apple instances and assigns each detection to one of three harvesting-oriented occlusion levels: non-occluded (NO), soft-occluded (SO), or hard-occluded (HO). Frequency-domain fusion (FAPEF), contextual modeling (TCG), and weighted multi-scale fusion (BiFPN) are integrated to improve robustness under branch and leaf occlusion, fruit overlap, weak boundaries, and illumination variation.

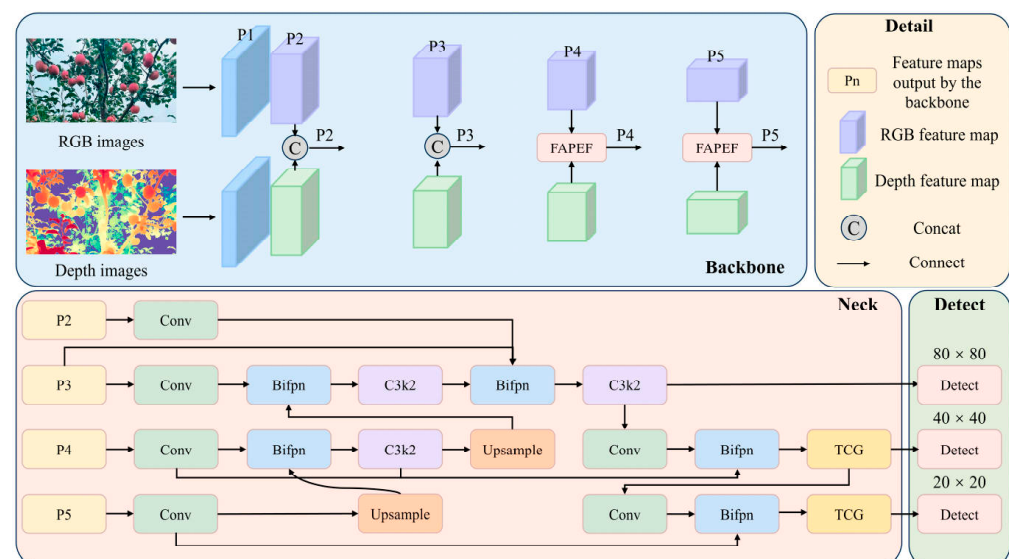


Figure 1. Overview of OAFBC-Net.

First, the network introduces the Fourier amplitude-phase enhancement fusion (FAPEF) module at the P4 and P5 feature maps. Each modal stream is first enhanced and spatially stabilized by DyT; FAPEF then re-transforms the stabilized features into the frequency domain, adaptively fuses the amplitude and principal-value phase spectra of RGB and depth, reconstructs the complex spectrum, and maps the fused representation back to the spatial domain through an orthonormal inverse frequency transform followed by DyT.

Second, the Triple Context Guided Block (TCG Block) is applied to the FAPEF outputs at P4 and P5 to strengthen the semantic association between local visible regions and surrounding contextual information. In occlusion scenes, visible apple regions are often incomplete; the TCG Block supplements missing semantic cues by modeling spatial relationships among apples, branches, leaves, adjacent fruits, and background regions, thereby improving discrimination among NO, SO, and HO categories.

Third, the network adopts a weighted bidirectional multi-scale fusion structure, BiFPN, in the neck to integrate P3, P4, and P5 features from different levels. BiFPN measures the importance of features at different scales through learnable weights and builds both top-

down and bottom-up information transfer paths, enabling sufficient fusion between shallow detail features and deep semantic features.

Overall, OAFBC-Net integrates FAPEF, the TCG Block, and BiFPN into a unified framework, enabling joint optimization of frequency-domain fusion, contextual semantic modeling, and multi-scale feature interaction. The network finally outputs the spatial location and occlusion level of each apple, providing a visual perception basis for target selection, approach-direction planning, and picking-strategy formulation in picking robots. Figure 2 presents the detailed backbone structure and key modules.

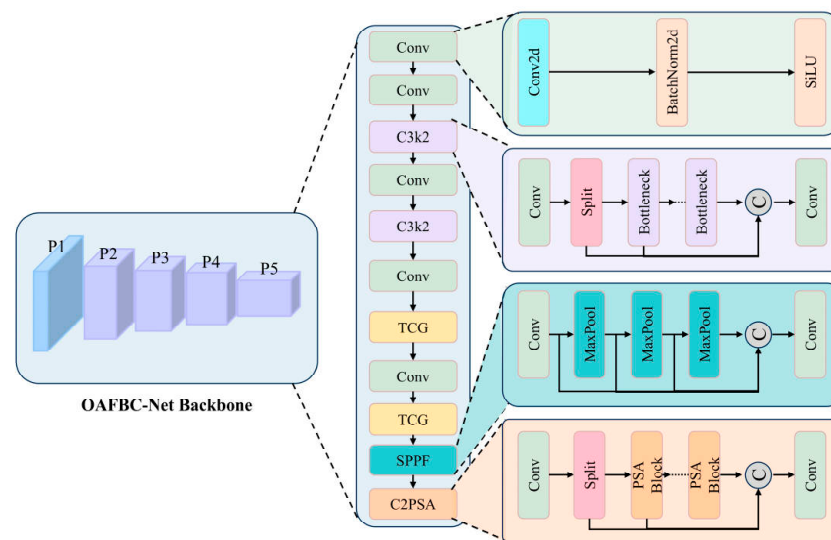


Figure 2. Detailed diagram of the backbone network structure of OAFBC-Net.

2.2. Fourier Amplitude-Phase Enhancement Fusion Module (FAPEF)

To strengthen cross-modal representation of occluded apples, the Fourier amplitude-phase enhancement fusion module (FAPEF) is embedded in a lightweight dual-branch RGB-depth detector inspired by UMIS-style frequency fusion [37]. Network implementation details, training settings, and the dual-modal baseline are given in Sections 3.2 and 3.4.

To fuse RGB and depth features more effectively, this paper designs the FAPEF module, which performs frequency-domain enhancement and cross-modal fusion at the high-level semantic stages of the backbone network. As shown in Figure 3, let X_{RGB}^s and X_{Depth}^s denote the real-valued feature tensors output by the RGB branch and the depth branch, respectively, at scale s . These tensors are defined in Equation (1) as

$$X_{RGB}^s, X_{Depth}^s \in \mathbb{R}^{B \times C_s \times H_s \times W_s} \quad (1)$$

where $\mathbb{R}$ denotes the set of real numbers; X_{RGB}^s is the RGB-branch feature tensor at scale s ; X_{Depth}^s is the depth-branch feature tensor at scale s ; $s \in \{4, 5\}$ denotes the P4 and P5 feature layers where FAPEF is deployed; B denotes the batch size; and C_s , H_s , and W_s denote the number of channels, height, and width of the feature map at scale s , respectively. Considering that the P3 layer has a higher resolution and larger computational cost, FAPEF is introduced only at P4 and P5, which contain richer semantic information and smaller spatial sizes. For P3, RGB-depth fusion is completed through lightweight feature concatenation and convolutional compression.

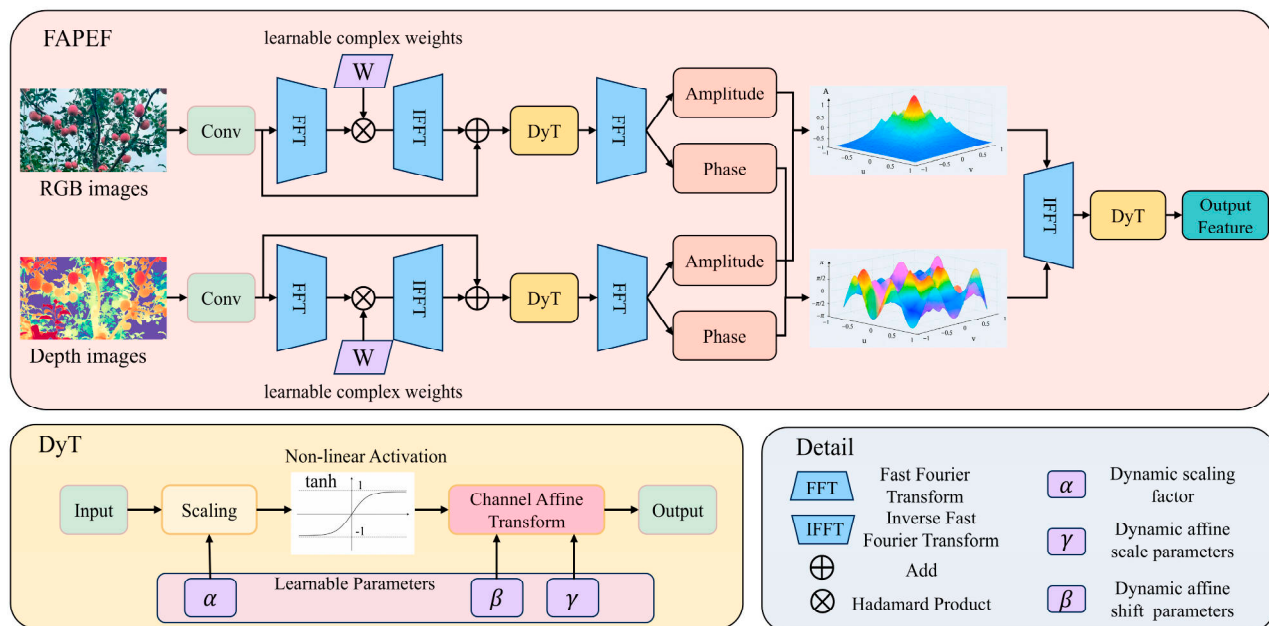


Figure 3. Structure of FAPEF.

Stage 1: Single-modal frequency enhancement. FAPEF first maps each modal feature into the frequency domain using an orthonormal two-dimensional (2-D) real fast Fourier transform (FFT; $\text{FFT}_{\text{ortho}}$, implemented as `rfft2` with `norm = 'ortho'`) and reconstructs the enhanced feature through the corresponding inverse fast Fourier transform (IFFT; $\text{IFFT}_{\text{ortho}}$). The single-modal enhancement process is given by Equations (2)–(4):

$$F_m^s = \text{FFT}_{\text{ortho}}(X_m^s) \quad (2)$$

$$\tilde{F}_m^s = F_m^s \odot W_m^s \quad (3)$$

$$\tilde{X}_m^s = \text{Re} \left\{ \text{IFFT}_{\text{ortho}} \left(\tilde{F}_m^s \right) \right\} + X_m^s \quad (4)$$

where $m \in \{\text{RGB}, \text{Depth}\}$, W_m^s is the learnable complex weight of modality m at scale s , $\odot$ denotes the Hadamard product, and $\text{FFT}_{\text{ortho}}(\cdot)$ and $\text{IFFT}_{\text{ortho}}(\cdot)$ denote the orthonormal FFT and IFFT operators, respectively. Through Equations (2) and (3), the model adaptively modulates the frequency response of each channel; Equation (4) reconstructs the spatial feature with a residual connection to preserve original modal information.

To stabilize the intermediate spatial activations, FAPEF employs Dynamic Tanh (DyT) [38]. Given an input feature tensor x , DyT first applies a learnable input scaling factor α to modulate the activation magnitude, then passes the scaled feature through the hyperbolic tangent function $\tanh(\cdot)$ to obtain a bounded nonlinear response, and finally applies channel-wise learnable scaling and bias through γ and β . The operator is defined in Equation (5) as

$$\text{DyT}(x) = \gamma \cdot \tanh(\alpha x) + \beta \quad (5)$$

where x denotes the input spatial feature tensor; α is a learnable scalar that controls the input sensitivity before the $\tanh$ nonlinearity; $\tanh(\alpha x)$ maps the scaled activation into the bounded interval $(-1, 1)$ and suppresses extreme values; γ is a channel-wise learnable scale vector that adjusts the amplitude of the bounded response; β is a channel-wise learnable bias vector that shifts the output activation; and the symbol $\cdot$ denotes element-wise multiplication when γ is applied across channels. Therefore, Equation (5)

can be interpreted as a bounded nonlinear remapping followed by channel-wise affine transformation. The enhanced feature is obtained by applying DyT in Equation (6):

$$\hat{X}_m^s = \text{DyT}(\tilde{X}_m^s) \quad (6)$$

where $\tilde{X}_m^s$ is the input feature to DyT at scale s . Since $|\tanh(\cdot)| \leq 1$, Equation (6) bounds the dynamic range of single-modal features before cross-modal fusion.

Stage 2: Cross-modal amplitude–phase fusion. After single-modal enhancement, FAPEF applies a second orthonormal FFT to the stabilized features in Equation (7):

$$F_m^s = \text{FFT}_{\text{ortho}}(\hat{X}_m^s) \quad (7)$$

The amplitude spectrum and principal-value phase spectrum are extracted in Equations (8) and (9), respectively:

$$A_m^s(u, v) = |F_m^s(u, v)| \quad (8)$$

$$P_m^s(u, v) = \text{atan2}(\text{Im}(F_m^s(u, v)), \text{Re}(F_m^s(u, v))) \quad (9)$$

where (u, v) denote frequency indices and $P_m^s(u, v) \in (-\pi, \pi]$ is the principal-value phase. Equation (8) reflects the energy of different frequency components; whereas, Equation (9) encodes spatial layout and structural position.

Cross-modal fusion is performed in Equations (10) and (11). Here, A_{fd}^s and P_{fd}^s denote the fused amplitude spectrum and the fused principal-value phase spectrum, respectively, at scale s in the frequency domain, where the subscript fd indicates frequency-domain fusion:

$$A_{fd}^s = \alpha_{RGB}^s \odot A_{RGB}^s + \alpha_{Depth}^s \odot A_{Depth}^s \quad (10)$$

$$P_{fd}^s = \beta_{RGB}^s \odot P_{RGB}^s + \beta_{Depth}^s \odot P_{Depth}^s \quad (11)$$

where A_{RGB}^s and A_{Depth}^s are the amplitude spectra of the RGB and depth branches obtained from Equation (8); P_{RGB}^s and P_{Depth}^s are the corresponding principal-value phase spectra obtained from Equation (9); A_{fd}^s is the cross-modally fused amplitude spectrum; P_{fd}^s is the cross-modally fused phase spectrum; and $\alpha_{RGB}^s, \alpha_{Depth}^s, \beta_{RGB}^s$, and β_{Depth}^s are channel-wise learnable fusion weights. Equations (10) and (11) allow FAPEF to adaptively balance RGB appearance and depth geometry at both amplitude and phase levels.

Numerical considerations in phase fusion. When amplitude and phase are fused separately, three numerical issues may arise in FAPEF at P4 and P5: (i) ill-conditioned phase extraction in Equation (9) when the amplitude in Equation (8) satisfies $|F_m^s(u, v)| \approx 0$; (ii) unbounded linear phase mixing in Equation (11), which may amplify high-frequency energy after inverse fast Fourier transform (IFFT) because $\exp(j\theta)$ is 2π -periodic; and (iii) magnitude scale drift across repeated fast Fourier transform (FFT)–inverse fast Fourier transform (IFFT) operations in depth.

Normalization and stabilization. In our implementation, the “normalization constraint” refers to orthonormal spectral transforms, principal-value phase parameterization, and bounded spatial remapping, rather than batch normalization on phase tensors. First, orthonormal fast Fourier transform (FFT)/inverse fast Fourier transform (IFFT) is used throughout FAPEF, satisfying Equation (12):

$$\text{IFFT}_{\text{ortho}}(\text{FFT}_{\text{ortho}}(X)) = X \quad (12)$$

Equation (12) preserves energy and limits transform-induced magnitude drift. Second, the principal-value phase in Equation (9) provides a consistent angular representation before fusion. Third, after Equations (10) and (11), the fused complex spectrum F_{fd}^s is reconstructed from the fused amplitude and phase spectra in Equation (13):

$$F_{fd}^s = A_{fd}^s \odot \exp(jP_{fd}^s) \quad (13)$$

and mapped back to the spatial domain in Equation (14):

$$\bar{X}_{fd}^s = \text{Re}\left\{\text{IFFT}_{\text{ortho}}\left(F_{fd}^s\right)\right\} \quad (14)$$

where F_{fd}^s denotes the fused complex spectrum in the frequency domain at scale s , and j is the imaginary unit. Finally, a second DyT module stabilizes the fused output in Equation (15):

$$X_{fd}^s = \text{DyT}\left(\bar{X}_{fd}^s\right) = \gamma^s \odot \tanh\left(\alpha^s \bar{X}_{fd}^s\right) + \beta^s, \quad s \in \{4, 5\} \quad (15)$$

Since $|\tanh(\cdot)| \leq 1$, Equation (15) bounds the fused spatial activations before they enter the TCG Block and BiFPN, thereby suppressing extreme values induced by phase mixing and inverse fast Fourier transform (IFFT) ringing. The same DyT operator defined in Equation (5) is used in both Equation (6) for single-modal enhancement and Equation (15) for fused-output stabilization.

Here, $X_{fd}^s \in \mathbb{R}^{B \times C_s \times H_s \times W_s}$ denotes the final RGB-depth fused feature at scale s given by Equation (15). In the proposed model, X_{fd}^4 and X_{fd}^5 are produced at the P4 and P5 FAPEF layers and serve as inputs to the subsequent TCG Block. After context enhancement, the P4 and P5 features are combined with the P3 features and then fed into BiFPN for further cross-scale weighted bidirectional feature fusion. In this way, FAPEF models frequency-domain complementarity between RGB and depth within the same scale, the TCG Block enhances local and contextual information, and BiFPN integrates multi-level features across scales. Together, these components provide a stable and discriminative multimodal feature representation for occluded apple detection in complex orchard environments.

2.3. BiFPN

To enhance the multi-scale detection ability for occluded apples in complex orchards, this paper adopts the Bidirectional feature pyramid network, BiFPN, in the Neck of OAFBC-Net [39]. BiFPN takes the RGB-depth fused P3, P4, and P5 features as inputs, where P3 represents shallow fused features, P4 represents middle-level fused features, and P5 represents high-level fused features. The structural comparison is shown in Figure 4. Compared with the traditional FPN-PAN network, which transfers information layer by layer along adjacent pyramid levels and usually adopts simple addition or concatenation for feature fusion, BiFPN introduces a bidirectional multi-path fusion mechanism. This mechanism supports cross-layer skip connections and repeated stacking, allowing low-level detail features and high-level semantic features to interact over multiple rounds. At the same time, BiFPN adopts learnable weighted feature fusion, which automatically adjusts the importance of features from different levels according to the task. This design makes fuller use of multi-scale information and significantly improves the model's perception ability for small apples, overlapping apples, and apples with different occlusion levels.

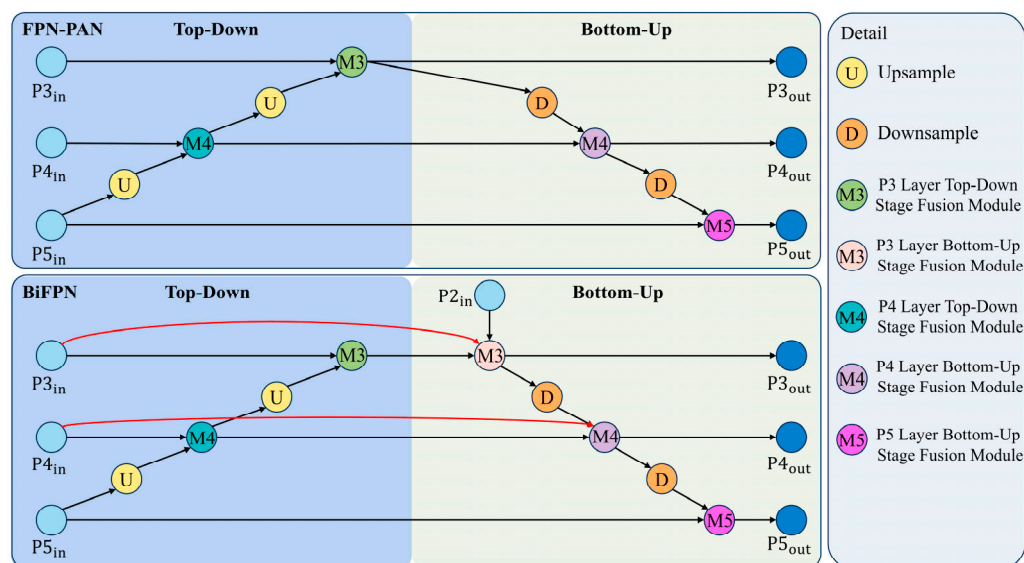


Figure 4. Comparison of FPN-PAN and BiFPN structures.

In the top-down feature transfer path, the high-level P5 feature first passes through a convolution layer to unify the number of channels to 256. It is then upsampled to the P4 resolution and fused with the P4 feature through weighted fusion. The fusion weights are learnable parameters, and normalization is used to ensure that the contributions of different scale features to the output are adaptively adjusted. The fused P4 feature is processed by the C3k2 module and then upsampled to the P3 resolution, where it is fused with the P3 feature to form an enhanced shallow feature representation. The fusion process is defined as follows:

$$P_4^{td} = \text{Conv} \left(\frac{\omega_1 \cdot P_4^{in} + \omega_2 \cdot \text{Upsample}(P_5^{in})}{\omega_1 + \omega_2 + \epsilon} \right), P_3^{td} = \text{Conv} \left(\frac{\omega_3 \cdot P_3^{in} + \omega_4 \cdot \text{Upsample}(P_4^{td})}{\omega_3 + \omega_4 + \epsilon} \right) \quad (16)$$

where $\text{Upsample}(\cdot)$ denotes the upsampling operation, $\text{Conv}(\cdot)$ denotes the convolution operation, ϵ is a small constant used to prevent numerical instability, and ω_i denotes learnable weights.

In the bottom-up path, the enhanced shallow P3 feature is first downsampled to the P4 resolution and combined with the previously fused P4 feature. The fused feature then passes through TCG modules to strengthen the middle-level representation. Next, the middle-level feature is downsampled to the P5 resolution, fused with the high-level P5 feature, and processed by TCG modules to generate the final P5 output feature. This process is defined as follows:

$$P_4^{out} = \text{Conv} \left(\frac{\omega'_1 \cdot P_4^{in} + \omega'_2 \cdot P_4^{td} + \omega'_3 \cdot \text{Downsample}(P_3^{td})}{\omega'_1 + \omega'_2 + \omega'_3 + \epsilon} \right) \quad (17)$$

$$P_5^{out} = \text{Conv} \left(\frac{\omega'_4 \cdot P_5^{in} + \omega'_5 \cdot P_5^{td} + \omega'_6 \cdot \text{Downsample}(P_4^{in})}{\omega'_4 + \omega'_5 + \omega'_6 + \epsilon} \right) \quad (18)$$

where $\text{Downsample}(\cdot)$ denotes the downsampling operation. During bidirectional fusion, the contribution of each input feature is dynamically adjusted by the learnable weight ω_i . The fusion formulation is defined as follows:

$$O = \sum_i \frac{\omega_i}{\epsilon + \sum_j \omega_j} \cdot I_i, \omega_i \geq 0 \quad (19)$$

where I_i denotes the input feature, O denotes the fused output feature, and ϵ is a small constant used to prevent division by zero. Through repeated bidirectional cyclic fusion, BiFPN enables the P3, P4, and P5 features to contain both shallow texture and edge information and deep semantic information, thereby significantly improving the model's discrimination ability for apples with different occlusion levels. The final fused features are sent to the detection head to predict the category and spatial location of apple targets. Combined with the FAPEF frequency-domain enhancement module and the TCG triple context guided module, the multi-scale weighted fusion of BiFPN provides strong feature support for robust occluded apple recognition by OAFBC-Net in complex orchard environments.

2.4. Triple Context Guided Block (TCG Block)

In multimodal object detection, targets are often difficult to distinguish from their surrounding environment because of scale variation, occlusion, and background texture interference. Standard convolution with only a local receptive field is insufficient to capture both fine-grained details and scene-level semantics. To address this issue, this paper proposes the Triple Context Guided Block (TCG Block), inspired by CGNet [40]. The core idea is to explicitly fuse three complementary types of context within the same basic operator, namely local context, surrounding context captured by dilated convolution, and global context captured by channel attention. These contextual cues are further integrated into the CSP structure through a global residual form, so that the block preserves efficient gradient flow in the backbone while capturing fine-grained edge information, spatially related environmental cues, and channel-level semantic recalibration.

Specifically, as shown in Figure 5, the TCG Block serves as the core operator for three-level context modeling. Given an input feature $X \in R^{B \times C \times H \times W}$, a 1×1 convolution is first used to compress the channel dimension and obtain X' . Two parallel depthwise separable convolution branches are then applied to the compressed feature. The local context branch $f_{loc}(\cdot)$ uses a standard 3×3 depthwise convolution to capture fine-grained details within each pixel's 3×3 neighborhood (the central pixel and its eight adjacent pixels). The surrounding context branch $f_{sur}(\cdot)$ uses a 3×3 depthwise dilated convolution with dilation rate $r_d = 2$, which expands the equivalent receptive field to 5×5 and models a larger neighborhood spatially related to the target. The outputs of the two branches are concatenated along the channel dimension and then fused by BatchNorm and the activation function $\sigma(\cdot)$, which denotes the Sigmoid activation function, to obtain the joint feature, as defined in Equation (20).

$$F_{joi} = \sigma(\text{BN}(\text{Concat}(f_{loc}(X'), f_{sur}(X')))) \quad (20)$$

After obtaining F_{joi} , which contains fused spatial context, the TCG Block applies the global context extractor FGlo to perform channel-level recalibration. This design makes spatial feature learning and channel feature enhancement complementary and relatively independent. Specifically, global average pooling is first used to form a channel descriptor. The descriptor is then passed through a two-layer MLP with a reduction ratio of $r = 16$, followed by a Sigmoid function, to obtain the channel attention vector. Finally, the channel-weighted feature is added to the original input through global residual learning to form the output of the core operator.

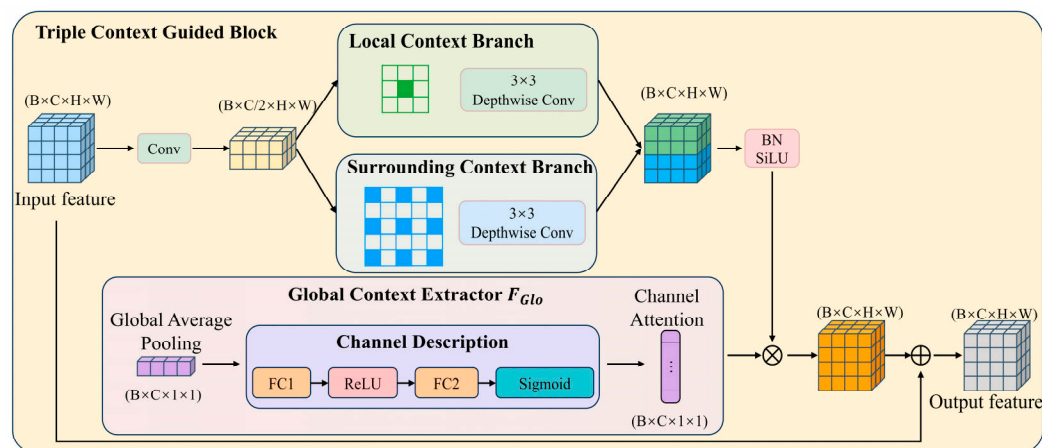


Figure 5. Structure of TCG Block.

At the network level, the TCG Block uses the CSP topology of C3k2 as its outer framework. The input feature map is first projected by a 1×1 convolution and split into two branches along the channel dimension. One branch sequentially passes through n three-context core operators described above. All original and intermediate outputs are then concatenated along the channel dimension and fused by another 1×1 convolution, which balances parameter efficiency and cross-layer gradient flow. Considering that shallow features mainly contain fine-grained textures and have weaker demand for large receptive fields and channel attention, while deep features have higher semantic abstraction and require broader context to distinguish occluded targets or targets similar to the background, this paper keeps the original C3k2 unit at the shallow P3 level and replaces it with the TCG Block only at the deeper P4 and P5 levels, where C3k-style nesting is enabled. In this way, the network fully aggregates local, surrounding, and global contexts at deeper layers with lower resolution and larger channel dimensions, thereby strengthening pixel-level discrimination in complex scenes while avoiding the extra computational cost caused by redundant dilated convolution at shallow layers.

2.5. Loss Function

During training, to ensure the fairness and comparability of experimental results across different models, the proposed OAFBC-Net and all comparison models are trained with the standard object detection loss function built into YOLOv11. This setting is used in all experiments in this paper, including performance comparisons among different algorithms, comparisons under RGB and depth input conditions, and ablation experiments on FAPEF, BiFPN, and TCG. The standard YOLOv11 detection loss mainly consists of three terms: distribution focal loss (dfl), object classification loss (cls), and bounding box localization loss (box). The total loss is defined as follows:

$$L_{total} = \lambda_1 L_{dfl} + \lambda_2 L_{cls} + \lambda_3 L_{box} \quad (21)$$

where L_{dfl} denotes the distribution focal loss, L_{cls} denotes the object classification loss, and L_{box} denotes the bounding box localization loss. λ_1 , λ_2 , and λ_3 denote the weight coefficients of the three loss terms, respectively. This paper adopts the default YOLOv11 weight setting. In actual training, these weights can also be adjusted according to specific detection tasks.

The distribution focal loss L_{dfl} is used to optimize the fine-grained regression of predicted box boundaries. Unlike direct regression of bounding box coordinates, DFL models the four coordinate offsets of a bounding box as discrete probability distributions.

This design enables the network to focus more on the discrete values near the ground-truth position and increase their predicted probabilities, thereby improving bounding box localization accuracy. Its calculation is defined as follows:

$$L_{dfl} = \sum_{i=0}^{K \times K} \sum_{p=0}^3 I_{ij}^{obj} \cdot DFL(s_i, s_{i+1}) \quad (22)$$

$$DFL(s_i, s_{i+1}) = -[(y_{i+1} - y) \log(s_i) + (y - y_i) \log(s_{i+1})] \quad (23)$$

where $K \times K$ denotes the number of grids on feature maps at different scales. When the input image size is 640×640 , YOLOv11 usually performs prediction at three scales, namely 80×80 , 40×40 , and 20×20 . p denotes the four boundary coordinates of the predicted box, and I_{ij}^{obj} indicates whether the j -th predicted target in the i -th grid is a positive sample. The distribution focal loss (DFL), which models each coordinate offset of a bounding box as a discrete probability distribution rather than as a direct continuous regression target, requires the hyperparameter reg_{max} to be set in advance. Here, reg_{max} denotes the number of discrete bins used to represent each coordinate offset in the distribution-based regression branch, and its default value is 16. Accordingly, the output channel number of the bounding box regression branch is $4 \times reg_{max}$, because four box coordinates (left, top, right, and bottom) are regressed independently. For a real continuous position y , its left and right adjacent integer positions are y_i and y_{i+1} , and the corresponding predicted probabilities are s_i and s_{i+1} . Through this probability distribution regression strategy, the model learns apple target boundaries more precisely, which is especially beneficial for improving the localization accuracy of softly occluded and heavily occluded apples.

The object classification loss L_{cls} adopts binary cross-entropy loss to measure the difference between predicted class probabilities and ground-truth class labels. The detection task in this paper contains three target categories, namely non-occluded apples (NO), softly occluded apples (SO), and heavily occluded apples (HO). Therefore, the classification branch needs to learn the feature differences among apples with different occlusion levels. The calculation is defined as follows:

$$L_{cls} = - \sum_{i=0}^{K \times K} I_{ij}^{obj} \sum_{c \in cls} \left[P_i^j(c) \log \hat{P}_i^j(c) + (1 - P_i^j(c)) \log (1 - \hat{P}_i^j(c)) \right] \quad (24)$$

where c denotes the target category, $P_i^j(c)$ denotes the ground-truth class label, and $\hat{P}_i^j(c)$ denotes the predicted probability that the target belongs to category c . Through this loss function, the model learns the differences among NO, SO, and HO apples in visible region size, occlusion degree, boundary completeness, and spatial structural features, thereby achieving occlusion-level classification and detection.

The bounding box localization loss L_{box} adopts the complete intersection over union (CIoU) loss function to measure the positional deviation between the predicted box and the ground-truth box. Compared with the ordinary intersection over union (IoU) loss, CIoU considers not only the overlap area between the predicted box and the ground-truth box, but also the center-point distance and aspect-ratio consistency constraints, which makes the bounding box regression process more stable. The calculation is defined as follows:

$$L_{box} = L_{CIoU} = 1 - \frac{B_{gt} \cap B_{pred}}{B_{gt} \cup B_{pred}} + \frac{\rho^2(B_{gt}, B_{pred})}{w_c^2 + h_c^2} + \alpha \nu \quad (25)$$

$$\alpha = \frac{\nu}{(1 - IoU) + \nu} \quad (26)$$

$$v = \frac{4}{\pi^2} \left(\arctan \frac{w_{gt}}{h_{gt}} - \arctan \frac{w_{pred}}{h_{pred}} \right)^2 \quad (27)$$

where B_{gt} denotes the ground-truth box, and B_{pred} denotes the predicted box. $\frac{B_{gt} \cap B_{pred}}{B_{gt} \cup B_{pred}}$ denotes the intersection over union (IoU) between the predicted box and the ground-truth box, as used in Equation (26). $\rho^2(B_{gt}, B_{pred})$ denotes the squared Euclidean distance between the center points of the ground-truth box and the predicted box. w_c and h_c denote the width and height of the smallest enclosing rectangle that can simultaneously cover the ground-truth box and the predicted box. w_{gt} , h_{gt} , w_{pred} , and h_{pred} denote the widths and heights of the ground-truth box and the predicted box, respectively. α is a balance coefficient, and v is used to measure the aspect-ratio consistency between the predicted box and the ground-truth box. Figure 6 shows the main geometric parameters involved in the CIoU loss, including the ground-truth box, the predicted box, the intersection and union regions, the center-point distance, and the smallest enclosing rectangle.

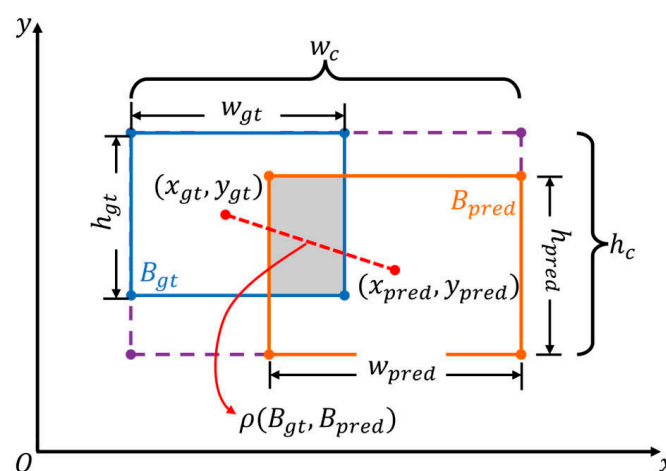


Figure 6. Geometric relationship of the CIoU loss function.

In summary, this paper jointly optimizes L_{dfl} , L_{cls} , and L_{box} to impose simultaneous constraints on apple target boundary positions, occlusion categories, and spatial localization accuracy. DFL improves the fine-grained regression ability of bounding box coordinates, BCE classification loss enhances the category discrimination ability for NO, SO, and HO apples, and CIoU localization loss improves the spatial matching degree between predicted boxes and ground-truth boxes. This standard loss function provides effective constraints for the stable training of OAFBC-Net and enables the model to achieve more accurate occluded apple detection in complex orchard environments.

3. Results

3.1. Dataset

The Apple dataset is an orchard RGB-depth image dataset collected at the First Station of the Taihang Mountain Road of Hebei Agricultural University in Shunping County, Baoding City, Hebei Province, China. Images are acquired using an Intel RealSense D455 camera (Intel Corporation, Santa Clara, CA, USA), which captures paired RGB images and depth maps at a resolution of 1280×720 pixels. Figure 7 shows representative visible-light and depth images from the dataset.

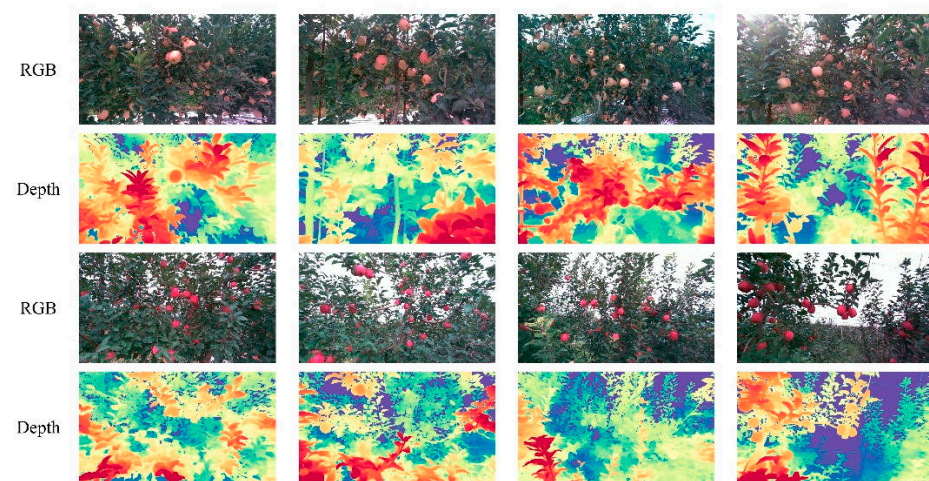


Figure 7. Some sample images from the Apple dataset.

This study follows the three-level occlusion classification standard for apple targets used in previous research, which is designed for harvesting passability and grasping feasibility. Apple targets are divided into three categories: No Occlusion (NO), Soft Occlusion (SO), and Hard Occlusion (HO) [41]. As shown in Figure 8, NO indicates that the apple target has a clear contour and a complete visible region, with little or no occlusion from branches, leaves, or adjacent fruits. Such targets can usually be directly used for visual localization and harvesting decisions. SO indicates that the apple is occluded by leaves, but its main body remains relatively distinguishable, and the end effector still has a high probability of successful picking after a slight push. HO indicates that the apple is severely occluded by branches, orchard structures, or adjacent fruits, which increases the difficulty of target localization and grasping decisions. Unlike conventional detection tasks that focus only on fruit localization, this classification standard further extends detection to a passability assessment task for motion planning and grasping decisions in harvesting robots. Therefore, the proposed model outputs not only the position coordinates of apple targets, but also their corresponding occlusion and rigidity-related category information, namely NO, SO, and HO, providing more task-oriented visual input for subsequent manipulator path planning, grasp pose selection, and obstacle avoidance decisions. The annotated instance categories are shown in Table 1.

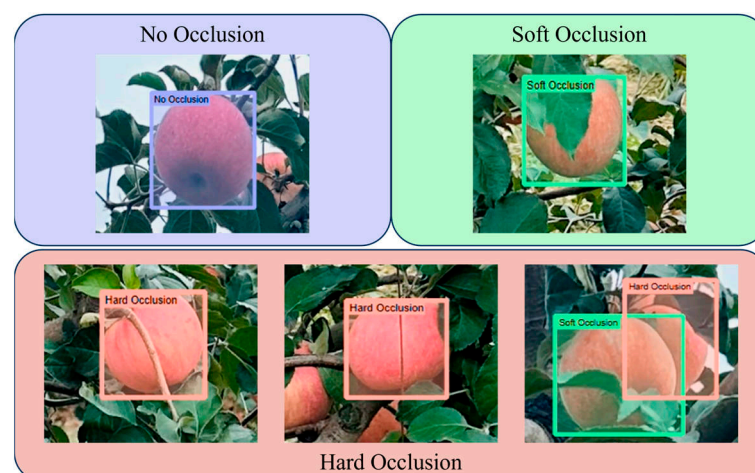


Figure 8. The three-tier apple occlusion classification taxonomy adopted in this study: No Occlusion (NO), Soft Occlusion (SO), and Hard Occlusion (HO).

Table 1. Number of apples with different occlusion types in the training, validation, and test sets of the Apple dataset.

Occlusion Type	Train	Val	Test	Sum
Soft Occlusion	40,849	5932	10,504	57,285
No Occlusion	22,148	3138	5720	31,006
Hard Occlusion	34,004	4563	8141	46,708

The dataset contains 3000 apple images, including 57,285 soft-occluded apples, 31,006 non-occluded apples, and 46,708 hard-occluded apples, for a total of 134,999 annotated apple instances. The images cover different illumination conditions, including front lighting, backlighting, and side lighting. The apples also vary in size and include small fruits, dense fruit-cluster occlusion, and visually similar instances, making automatic detection and recognition challenging.

Because the depth maps contain noise and missing regions, the original depth maps are filtered and optimized before model training. The processing operations include bilateral filtering, hole filling, and median filtering, which improve spatial continuity in the depth data. For the main comparison and ablation experiments, the Apple dataset is divided into training, validation, and test sets at a ratio of 7:1:2. To avoid data leakage, the split is conducted based on acquisition batches, tree IDs, and image groups rather than random splitting at the single-image level. This strategy prevents images from the same acquisition location or continuous shooting sequence from appearing simultaneously in the training, validation, and test sets.

To further evaluate stability under different data splits, five-fold cross-validation is conducted based on the fixed-split experiments. The original test set remains unchanged throughout cross-validation to preserve its independence. The original training and validation sets are merged into a development dataset and then divided into five folds using the same leakage-prevention principle as the main experiments. In each round, four folds are used for training and one fold is used for validation. After training each fold, the best-performing weights on the current validation set are evaluated on the same independent test set. The detailed results are presented in Section 3.6.

To analyze the spatial distribution and scale characteristics of the Apple dataset, this study conducts a quantitative statistical analysis of the annotated bounding boxes. As shown in Figure 9a, the normalized center coordinates of apple targets cover most regions of the image, with higher density near the image center. This distribution does not show an extreme spatial bias, although apple targets appear more frequently within the main field of view because of orchard row structure, camera viewpoint, and shooting distance. Figure 9b shows the normalized distribution of bounding box width and height. Most targets are concentrated within a small-scale range, with both width and height mainly below 0.10 of the image size. A positive correlation is observed between width and height, indicating that most annotated apple boxes maintain relatively stable object proportions. These characteristics show that small-scale and occluded targets account for a large proportion of the dataset, increasing the difficulty of localization and occlusion-level classification. Therefore, the dataset distribution supports the use of RGB-depth fusion and feature enhancement modules from the data perspective.

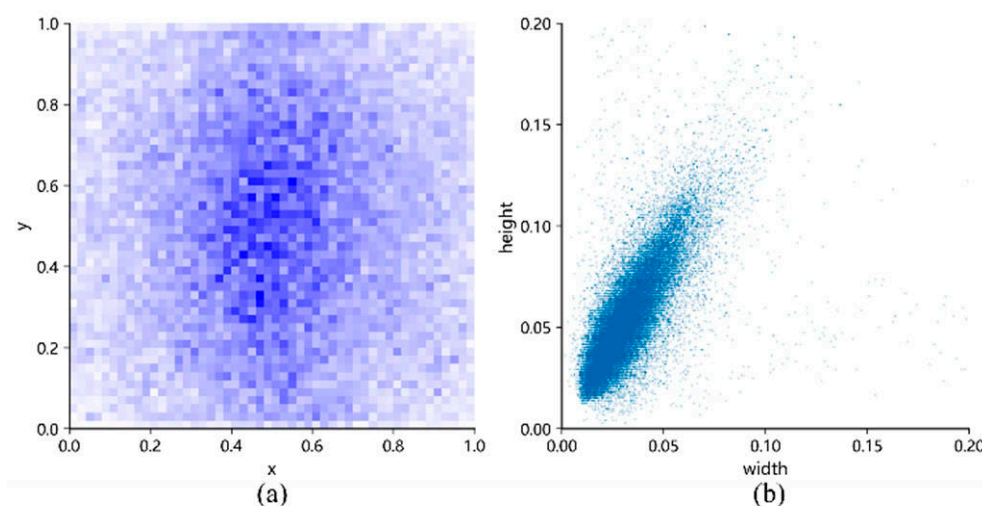


Figure 9. Visualization and statistical analysis of dataset characteristics. Darker blue regions indicate higher distribution density, and lighter blue regions indicate lower distribution density. (a) Spatial distribution of bounding box center coordinates. (b) Scale distribution of normalized bounding box width and height.

3.2. Experimental Platform and Parameters

To evaluate OAFBC-Net for apple occlusion classification and detection, this study conducts comparative and ablation experiments on the Apple dataset. Tables 2 and 3 list the hardware, software, and training parameters, which support the reproducibility and comparability of the experiments.

Table 2. Experimental environment configuration.

Environment	Parameters
GPU	NVIDIA GeForce RTX 4070 Laptop GPU (NVIDIA Corporation, Santa Clara, CA, USA)
CPU	Intel(R) Core(TM) i9-14900HX (2.20 GHz) (Intel Corporation, Santa Clara, CA, USA)
GPU memory	8 GB
Operating system	Windows 11
Language	Python 3.10.19
Framework	PyTorch 2.8.0
CUDA version	CUDA 12.6

Table 3. Training parameter settings.

Parameters	Setup
Epochs	300
Input image size (pixels)	640
Batch size	8
Workers	8
Learning rate	0.01
Momentum	0.937
Weight decay	0.0005
Optimizer	SGD

To support experimental reproducibility, the random seed is fixed during model training. Specifically, the random seeds of Python random, NumPy (version 2.2.6), and PyTorch are fixed, and all models use the same data split. Unless otherwise specified, the main comparative and ablation experiments use seed = 0. To verify that the performance improvement is not tied to a specific initialization, additional experiments are repeated with seed = 42 and seed = 2026.

3.3. Evaluation Metrics

Detection accuracy is evaluated using Precision (P), Recall (R), class-wise average Precision at IoU = 0.50 (AP50), and macro-averaged mAP50 and mAP50-95 over the three occlusion classes. Class-wise metrics are reported for Soft Occlusion (SO), No Occlusion (NO), and Hard Occlusion (HO). Each AP50 value is the area under the Precision-Recall curve for a given class, with a prediction counted as correct when $\text{IoU} \geq 0.50$. Macro mAP50 is the arithmetic mean of the three class-wise AP50 values, and macro mAP50-95 denotes the mean under the default validation protocol of the detection framework. Model complexity is reported as parameter count and giga floating-point operations (GFLOPs) at an input resolution of 640×640 pixels. Inference efficiency is additionally evaluated using per-image preprocessing time, inference time, postprocessing time, total latency, and frames per second (FPS), as reported in Section 3.7.

During evaluation, predicted boxes and ground-truth boxes of the same class were matched in a one-to-one manner. A matched prediction was counted as a True Positive (TP) when $\text{IoU} \geq 0.50$. An unmatched prediction was counted as a False Positive (FP), and an unmatched ground-truth box was counted as a False Negative (FN). Precision, Recall, and class-wise AP50 were computed as follows:

$$P = \frac{TP}{TP + FP} \quad (28)$$

$$R = \frac{TP}{TP + FN} \quad (29)$$

$$mAP50 = \frac{1}{N} \sum_{i=1}^N (P(R) dR)_i \quad (30)$$

where $P(R)$ denotes the Precision-Recall curve of class c , and $N = 3$ is the number of occlusion classes (NO, SO, and HO). For comparative experiments in Table 4, $AP50_{SO}$, $AP50_{NO}$, and $AP50_{HO}$ are reported separately to reflect detection performance under different occlusion levels. For ablation experiments in Table 5, overall mAP50 and mAP50-95 are reported together with class-wise AP50 values. For multi-seed and cross-validation experiments in Tables 6 and 7, Precision, Recall, mAP50, and mAP50-95 are reported. All models were trained and tested under the same experimental settings, input resolution, and data split, so that accuracy, GFLOPs, and parameter counts are directly comparable.

Table 4. Performance comparison between OAFBC-Net and different algorithms.

Method	Modality	AP50 (%)			FLOPs (G)	Params (M)
		SO	NO	HO		
YOLOv5	RGB-only	63.4	81.1	44.3	5.8	2.2
YOLOv8	RGB-only	63.4	80.8	44.9	6.8	2.7
YOLOv10	RGB-only	63.6	81.1	45.7	6.5	2.3
YOLOv11	RGB-only	63.7	81.3	46.6	6.3	2.6

Table 4. Cont.

Method	Modality	AP50 (%)			FLOPs (G)	Params (M)
		SO	NO	HO		
SSD	RGB-only	59.6	77.0	43.9	30.5	24.0
Faster R-CNN	RGB-only	67.3	82.9	47.8	208.0	41.4
RetinaNet	RGB-only	57.8	80.8	41.0	156.0	19.8
YOLOv5	depth-only	35.4	74.9	22.9	5.8	2.2
YOLOv8	depth-only	31.4	71.7	20.2	6.8	2.7
YOLOv10	depth-only	32.2	74.1	21.2	6.5	2.3
YOLOv11	depth-only	34.3	74.3	20.5	6.3	2.6
SSD	depth-only	26.1	67.1	14.9	30.5	24.0
Faster R-CNN	depth-only	41.8	78.0	28.6	208.0	41.4
RetinaNet	depth-only	27.4	69.2	14.2	156.0	19.8
YOLOv5-RGBD	RGB + depth	65.2	82.3	45.5	8.3	4.4
YOLOv8-RGBD	RGB + depth	65.2	82.4	45.7	13.2	5.3
YOLOv10-RGBD	RGB + depth	65.4	82.2	45.9	11.8	4.7
YOLOv11-RGBD	RGB + depth	65.0	82.4	46.9	11.5	5.0
OAFBC-Net	RGB + depth	72.1	87.3	57.9	10.2	3.6

Note: FLOPs (G) denotes giga floating-point operations; Params (M) denotes model parameters in millions (640×640 input). Bold values indicate the best result in each column across the entire table: the highest AP50 for SO, NO, and HO, and the lowest FLOPs and Params.

Table 5. Ablation experiment results.

Model	FAPEF	BiFPN	TCG	mAP50 (%)	mAP50-95 (%)	AP50 (%)			FLOPs (G)	Params (M)
						SO	NO	HO		
1				64.8	47.9	65.0	82.4	46.9	11.5	5.0
2	✓			70.5	53.1	70.3	86.6	54.7	10.6	4.7
3		✓		66.4	48.7	67.0	84.5	47.8	11.5	4.3
4			✓	66.0	49.3	65.7	83.1	49.3	10.4	4.2
5	✓	✓		71.7	54.0	71.8	87.1	56.4	11.3	4.2
6	✓		✓	71.2	53.8	71.9	87.1	54.6	10.2	4.1
7		✓	✓	70.4	52.8	70.5	86.4	54.4	10.4	3.7
8	✓	✓	✓	72.4	55.2	72.1	87.3	57.9	10.2	3.6

Note: Model 1 is the dual-modal baseline without FAPEF, BiFPN, or TCG; ✓ indicates that the corresponding module is included in the model; Models 2–4 add one module each; Models 5–7 add pairwise combinations; Model 8 is the complete OAFBC-Net. mAP50 and mAP50-95 are overall metrics computed using the default YOLOv11 validation protocol; AP50 columns report class-wise results at IoU = 0.50.

Table 6. Multi-seed experimental results.

Model	Seed	Precision (%)	Recall (%)	mAP50 (%)	mAP50-95 (%)
YOLOv11-RGBD	0	59.2	68.6	65.8	47.9
	42	58.1	68.7	64.7	48.1
	2026	57.7	69.7	66	48.1
	mean ± std	58.3 ± 0.8	69.0 ± 0.6	65.5 ± 0.6	48.0 ± 0.1
OAFBC-Net	0	64.8	71.7	72.4	55.2
	42	65.5	70.4	72.4	55.1
	2026	64.2	70.5	71.5	54.6
	mean ± std	64.8 ± 0.5	70.9 ± 0.7	72.1 ± 0.5	55.0 ± 0.3

Table 7. Five-fold cross-validation experimental results.

Model	Fold	Precision (%)	Recall (%)	mAP50 (%)	mAP50-95 (%)
YOLOv11-RGBD	Fold 1	57.7	66.8	64.4	47.5
	Fold 2	61.5	68.5	68.7	49.7
	Fold 3	58.3	69.3	67.6	49.0
	Fold 4	55.9	68.2	64.4	46.5
	Fold 5	57.5	67.8	65.0	47.3
	mean $\pm$ std	58.2 $\pm$ 2.1	68.1 $\pm$ 0.9	66.0 $\pm$ 2.0	48.0 $\pm$ 1.3
OAFBC-Net	Fold 1	64.1	70.0	71.6	54.0
	Fold 2	64.9	70.1	72.6	54.0
	Fold 3	63.0	70.0	70.7	52.5
	Fold 4	62.6	69.7	70.9	53.0
	Fold 5	63.8	69.7	71.1	52.7
	mean $\pm$ std	63.7 $\pm$ 0.9	69.9 $\pm$ 0.2	71.4 $\pm$ 0.8	53.2 $\pm$ 0.7

3.4. Comparison Experiment

The comparison models are grouped into three categories. The first category includes lightweight YOLO-series detectors, namely YOLOv5, YOLOv8, YOLOv10, and YOLOv11. The second category includes classical detectors, including SSD, RetinaNet, and Faster R-CNN. The third category includes multimodal YOLO-based baselines. To ensure fair comparison under dual-modal input conditions, this study constructs RGB-depth fusion versions of YOLOv5, YOLOv8, YOLOv10, and YOLOv11, denoted as YOLOv5-RGBD, YOLOv8-RGBD, YOLOv10-RGBD, and YOLOv11-RGBD, respectively. These models are self-implemented RGB-depth baselines based on the corresponding YOLO architectures, not official YOLO variants. OAFBC-Net is evaluated as the proposed RGB-depth detection method. Experiments are conducted under three input settings: RGB-only, depth-only, and RGB + depth fusion. All models are compared under the same experimental conditions.

As shown in Table 4, the detection performance of different models varies clearly across input modalities. For single-modal detection methods, the Only-RGB models consistently outperformed the depth-only models. This is mainly because RGB images contain rich color, texture, and edge information, which more effectively represents the appearance of apple targets. In contrast, depth images provide spatial structural cues, but their texture details and semantic representation are relatively limited when used alone, leading to a clear decline in detection performance. Under the Only-RGB setting, Faster R-CNN achieved 67.3%, 82.9%, and 47.8% on SO, NO, and HO, respectively, showing strong detection accuracy. However, its computational cost and parameter size reach 208.0 G and 41.4 M, respectively, which indicates high model complexity and limits practical deployment. By comparison, the YOLO-series models have lower FLOPs and fewer parameters, but their detection accuracy still has room for improvement.

A further comparison under the depth-only setting shows that all models suffer different degrees of performance degradation. For example, YOLOv10 achieved 63.6%, 81.1%, and 45.7% on SO, NO, and HO under the Only-RGB setting, but the corresponding values decrease to 32.2%, 74.1%, and 21.2% under the depth-only setting. This result indicates that depth information alone cannot fully describe the appearance features of apple targets. In complex scenes or cases with unclear object boundaries, the discriminative ability of the model is therefore limited. Thus, relying on a single modality makes it difficult to obtain stable and robust detection performance.

Compared with single-modal methods, RGB + depth multimodal fusion methods achieved better overall performance. YOLOv5-RGBD, YOLOv8-RGBD, YOLOv10-RGBD, and YOLOv11-RGBD all improve over their corresponding RGB-only models on SO, NO,

and HO, which indicates that the texture information from RGB images and the spatial structural information from depth images are strongly complementary. By fusing features from the two modalities, the models obtain a more complete target representation and thus improve detection performance.

Among RGB-depth methods under identical training settings, OAFBC-Net achieved the highest class-wise AP50 on SO, NO, and HO (72.1%, 87.3%, and 57.9%, respectively). Relative to the strongest dual-modal baseline (YOLOv11-RGBD, with AP50 values of 65.0%, 82.4%, and 46.9% for SO, NO, and HO), OAFBC-Net improved AP50_{HO} by 11.0 percentage points while reducing parameters from 5.0 M to 3.6 M. Faster R-CNN (RGB-only) is reported only as a single-modal reference because its input setting and architecture differ from the proposed RGB-depth framework.

In addition, among the multimodal detection models, OAFBC-Net has only 3.6 M parameters, fewer than YOLOv5-RGBD, YOLOv8-RGBD, YOLOv10-RGBD, and YOLOv11-RGBD. This shows that the proposed method improves detection accuracy while maintaining a lightweight model structure. Although its FLOPs are slightly higher than those of YOLOv5-RGBD, its detection accuracy improves substantially. In particular, OAFBC-Net improves AP50_{HO} by 12.4% over YOLOv5-RGBD, which indicates that it fuses RGB and depth information more effectively and strengthens the recognition ability for challenging targets.

Overall, the proposed OAFBC-Net fully exploits the texture details of RGB images and the spatial structural information of depth images, achieving the best performance across all detection categories. At the same time, the method keeps both parameter size and computational complexity at a low level, balancing detection accuracy and model lightweightness. These results demonstrate its strong robustness and practical value for occluded apple detection in complex orchard environments.

3.5. Ablation Experiment

To comprehensively evaluate the effectiveness and necessity of the proposed components, we conduct extensive ablation experiments on FAPEF, BiFPN, and TCG under identical training settings. The eight ablation configurations and their results are summarized in Table 5, which reports overall mAP50 and mAP50-95, class-wise AP50 values for Soft Occlusion (SO), No Occlusion (NO), and Hard Occlusion (HO), as well as FLOPs and parameter counts.

As shown in Table 5, the baseline model (Model 1) achieves the lowest overall and class-wise performance among all ablation configurations, with mAP50 of 65.8%, mAP50-95 of 47.9%, and AP50 values of 65.0%, 82.4%, and 46.9% for SO, NO, and HO, respectively. Compared with Model 1, introducing FAPEF alone (Model 2) improves mAP50 from 65.8% to 70.5% and mAP50-95 from 47.9% to 53.1%, while AP50 increases to 70.3%, 86.6%, and 54.7% for SO, NO, and HO, respectively. The parameter count decreases from 5.0 M to 4.7 M. This indicates that replacing the original concatenation-based fusion at the P4 and P5 stages with frequency-domain RGB-depth fusion enables the model to integrate cross-modal information more effectively and yields the largest single-module gain, particularly for HO detection (+7.8 percentage points in AP50_{HO}).

After introducing BiFPN alone (Model 3), mAP50 and mAP50-95 increase to 66.4% and 48.7%, respectively, while AP50 values become 67.0%, 84.5%, and 47.8% for SO, NO, and HO. The parameter count further decreases to 4.3 M. These results show that the bidirectional multi-scale fusion structure improves multi-scale feature representation and maintains computational efficiency, although its standalone contribution to overall accuracy is more moderate than that of FAPEF. After introducing TCG alone (Model 4), mAP50 and mAP50-95 reach 66.0% and 49.3%, respectively, with AP50 values of 65.7%, 83.1%, and

49.3% for SO, NO, and HO, while the parameter count decreases to 4.2 M. This suggests that TCG enhances contextual modeling and semantic compensation under occlusion, with a relatively more evident benefit on HO cases (+2.4 percentage points in AP50_HO over Model 1), even though its overall gain when used alone remains limited.

Furthermore, pairwise combinations of the proposed modules (Models 5–7) demonstrate clear synergistic effects. Model 5 (FAPEF + BiFPN) achieves mAP50 of 71.7% and mAP50-95 of 54.0%, with AP50 values of 71.8%, 87.1%, and 56.4% for SO, NO, and HO. Model 6 (FAPEF + TCG) reaches mAP50 of 71.2% and mAP50-95 of 53.8%, with AP50 values of 71.9%, 87.1%, and 54.6% for SO, NO, and HO. Model 7 (BiFPN + TCG) obtains mAP50 of 70.4% and mAP50-95 of 52.8%, with AP50 values of 70.5%, 86.4%, and 54.4% for SO, NO, and HO. All three pairwise variants outperform the corresponding single-module configurations, confirming that the proposed components provide complementary rather than redundant benefits. The complete model (Model 8) achieves the best performance across all reported metrics, with mAP50 of 72.4%, mAP50-95 of 55.2%, and AP50 values of 72.1%, 87.3%, and 57.9% for SO, NO, and HO, respectively, while using the lowest parameter count (3.6 M) and 10.2 GFLOPs among all eight configurations.

These results demonstrate that each component contributes not only independently but also complementarily to the final performance improvement. Relative to Model 1, the full model improves mAP50 and mAP50-95 by 6.6 and 7.3 percentage points, respectively, and increases AP50_HO by 11.0 percentage points. FAPEF enhances cross-modal frequency-domain fusion, BiFPN strengthens multi-scale feature interaction, and TCG improves contextual reasoning under occlusion. Importantly, the consistent performance drop across SO, NO, and HO categories when any module is removed validates the necessity of all three components for robust occlusion-aware detection.

To further substantiate the necessity of each proposed module, we compare the full model (Model 8) with its ablated variants in Table 5. When FAPEF is removed (Model 7), mAP50 decreases from 72.4% to 70.4% and mAP50-95 from 55.2% to 52.8%, with AP50_HO decreasing from 57.9% to 54.4%, indicating the importance of frequency-domain cross-modal fusion. When BiFPN is removed (Model 6), mAP50 decreases to 71.2% and mAP50-95 to 53.8%, showing that BiFPN contributes to multi-scale feature representation and overall localization quality. When TCG is removed (Model 5), mAP50 decreases to 71.7% and mAP50-95 to 54.0%, while AP50_HO decreases from 57.9% to 56.4%, indicating that TCG further strengthens contextual reasoning under severe occlusion. This degradation is consistently observed across SO, NO, and HO categories rather than being limited to a single occlusion subset. Therefore, the ablation results validate the importance of all proposed components and demonstrate their complementary roles in improving overall detection performance.

In addition, Figure 10 presents the normalized visualization of the ablation results. All data in the figure are normalized to the range of 0 to 1. For detection performance metrics, including P, R, mAP50, and mAP50-95, a value closer to 1 indicates better model performance. For the Params metric, a value closer to 0 indicates fewer model parameters. Model 1 to Model 8 in the figure correspond to the eight ablation configurations listed in Table 5. Model 8 shows the largest coverage in the accuracy-related dimensions, including P, R, mAP50, and mAP50-95, while maintaining the smallest parameter scale. This indicates that the final model achieves the most balanced performance among all ablation settings. Therefore, the combination of FAPEF, BiFPN, and TCG effectively improves detection accuracy and contributes to the construction of a lightweight detection model.

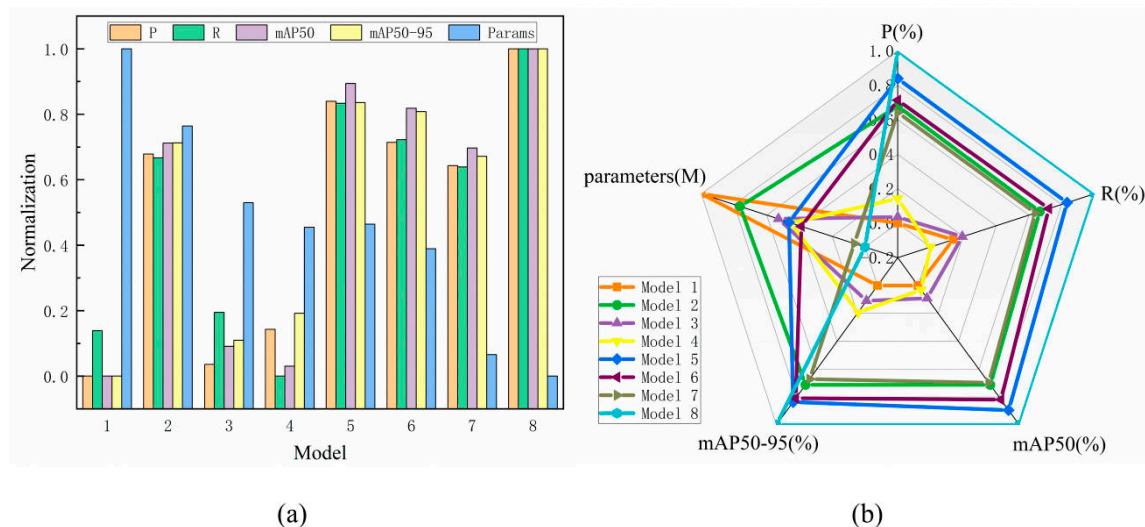


Figure 10. Normalization and performance analysis of ablation experiment indicators. (a) Histogram of normalized indicators. (b) Performance comparison of ablation variants.

3.6. Reproducibility and Cross-Validation Analysis

Although the preceding comparative experiments and ablation experiments are conducted under fixed data splits and a fixed random seed, the training results of deep learning detection models may still be affected by random initialization, the randomness of data augmentation, and the partitioning strategy for the training and validation sets. To further verify the reliability of the performance improvement achieved by the proposed model, this paper supplements the main comparative experiments and ablation experiments with multi-seed repeated experiments and five-fold cross-validation experiments. The multi-seed experiments are used to analyze the effect of different random initializations on model performance under unchanged data splits, while the five-fold cross-validation experiments are used to analyze the effect of different training and validation splits on model performance under a fixed random seed.

Considering that repeated training of all models would significantly increase the experimental cost, this paper selects two representative models for stability validation, namely YOLOv11-RGBD and OAFBC-Net. YOLOv11-RGBD is the RGB-D dual-modal fusion baseline model constructed in this paper based on the YOLOv11 architecture, and OAFBC-Net is the proposed occluded apple detection model. Through repeated experiments on these two models, the influence of RGB-D dual-modal information and the proposed feature enhancement modules on the stability of detection performance can be further analyzed.

To verify whether the performance improvement of OAFBC-Net is caused by the accidental effect of a specific random seed, this paper uses multiple random seeds to conduct repeated model training. In this experiment, the training, validation, and test splits remain unchanged, and only the random seed in the training process is changed. Specifically, three independent training runs are conducted with seed = 0, seed = 42, and seed = 2026.

During each training run, the random seeds of Python random, NumPy, and PyTorch are set to the same value. The experimental conditions, including input image size, number of training epochs, optimizer, learning rate strategy, batch size, data augmentation method, and evaluation metrics, are kept consistent. In this way, the differences among experimental results can be attributed mainly to random initialization and training randomness, rather than differences in experimental configurations.

This paper records the Precision, Recall, mAP50, and mAP50-95 of the two models under different random seeds, and further calculates the mean and standard deviation of the results obtained from the three random seeds. The mean reflects the overall detection performance of the model under different random initialization conditions, while the standard deviation measures the fluctuation of the training results. If a model maintains high detection accuracy with a small standard deviation under multiple random seeds, its performance improvement can be considered reproducible rather than an accidental result caused by a specific random seed.

As shown in Table 6, OAFBC-Net exhibits good stability in Precision, Recall, mAP50, and mAP50-95 under three different random seeds, with average values of $64.8 \pm 0.5\%$, $70.9 \pm 0.7\%$, $72.1 \pm 0.5\%$, and $55.0 \pm 0.3\%$, respectively. Compared with YOLOv11-RGBD, OAFBC-Net improves the average Precision, Recall, mAP50, and mAP50-95, among which mAP50 and mAP50-95 increase by 6.6 and 7.0 percentage points, respectively. This indicates that the proposed method achieves better detection performance under different random initialization conditions. In terms of standard deviation, all metrics of OAFBC-Net show small fluctuations. In particular, the standard deviations of mAP50 and mAP50-95 are 0.5% and 0.3%, respectively, indicating that the model has good consistency in overall detection accuracy and localization performance under multiple IoU thresholds. By contrast, the mAP50 and mAP50-95 of YOLOv11-RGBD are $65.5 \pm 0.6\%$ and $48.0 \pm 0.1\%$, respectively, suggesting that although YOLOv11-RGBD also shows a certain degree of stability, its overall detection performance is clearly lower than that of OAFBC-Net. Notably, while improving detection accuracy, OAFBC-Net still maintains low standard deviations in Precision, Recall, and mAP50, which indicates that the proposed RGB-D feature fusion and occlusion feature enhancement strategies do not increase the sensitivity of the model to random initialization. Overall, the multi-seed experimental results show that the performance improvement of OAFBC-Net is not accidental, but is stable, reproducible, and consistently advantageous under different random seeds.

Considering that the test set should remain as independent as possible, this paper keeps the original independent test set unchanged during five-fold cross-validation and only repartitions the development dataset formed by the original training and validation sets. Specifically, the original training set and validation set are first merged to form the development dataset, which is then divided into five non-overlapping subsets. In each experimental round, four subsets are selected as the training set, and the remaining subset is used as the validation set. After model training is completed, the model weights with the best performance on the current validation set are selected and evaluated on the independent test set, which remains unchanged throughout the process. This procedure is repeated five times until each subset is used once as the validation set.

To analyze the influence of data partitioning on model performance as independently as possible, seed = 0 is uniformly adopted in the five-fold cross-validation experiments. In addition, the input image size, number of training epochs, optimizer, learning rate strategy, batch size, data augmentation method, and evaluation metrics are kept consistent with those in the main comparative experiments for all folds.

As shown in Table 7, OAFBC-Net outperforms YOLOv11-RGBD in Precision, Recall, mAP50, and mAP50-95 in the five-fold cross-validation experiments. The average gains are 5.5, 1.8, 5.4, and 5.2 percentage points, respectively. OAFBC-Net also shows smaller standard deviations than YOLOv11-RGBD across all metrics. In particular, the standard deviations of Recall, mAP50, and mAP50-95 are 0.2%, 0.8%, and 0.7%, respectively, which are lower than the corresponding values of 0.9%, 2.0%, and 1.3% for YOLOv11-RGBD. These results indicate lower sensitivity to data partitioning and better reproducibility within the tested dataset and protocol.

3.7. Inference Efficiency Analysis

Parameter count and GFLOPs alone cannot fully support operational viability claims for OAFBC-Net, because FAPEF performs multiple two-dimensional fast Fourier transform (FFT) and inverse fast Fourier transform (IFFT) operations inside deep feature tensor channels. Depending on the deployment platform, such frequency-domain operations may introduce computational bottlenecks or memory-transfer latency. To address this concern, this section reports measured inference efficiency for all eight ablation configurations in Table 5 under an identical runtime protocol. The goal is to verify whether FAPEF prevents real-time inference and to provide empirical support for the lightweight design of OAFBC-Net.

All latency tests were conducted on the experimental platform described in Section 3.2 and Table 2, using an NVIDIA GeForce RTX 4070 Laptop GPU (8 GB), Intel Core i9-14900HX CPU, Windows 11, Python 3.10.19, PyTorch 2.8.0, and CUDA 12.6. At present, edge hardware such as NVIDIA Jetson devices are not available in this study. Therefore, the reported results should be interpreted as desktop/laptop GPU measurements rather than as direct evidence of embedded orchard deployment. Validation on edge platforms, including Jetson-based field perception systems, will be conducted in future work.

The eight ablation models in Table 5 were selected because they share the same RGB-depth input setting, training protocol, and overall detector framework, which enables controlled analysis of the runtime impact of FAPEF, BiFPN, and the TCG Block. Model 1 serves as the dual-modal baseline without any proposed module. Models 2–4 introduce FAPEF, BiFPN, and the TCG Block individually. Models 5–7 represent pairwise combinations, and Model 8 is the complete OAFBC-Net. This design allows direct comparison between Model 8 and Model 7 to isolate the runtime effect of FAPEF, while Model 1 and Model 2 further quantify the overhead of frequency-domain fusion alone.

For each model, inference efficiency was measured with paired RGB-depth inputs at a resolution of 640×640 pixels and batch size = 1. Each model was warmed up for 50 iterations and then evaluated over 1000 consecutive runs. The reported preprocessing time, inference time, and postprocessing time denote the mean per-image latency. Total latency was computed as the sum of these three terms, and FPS was calculated as $1000/\text{Total latency}$. All tests used the same trained weights as in the ablation experiments in Section 3.5.

Table 8 summarizes the measured runtime results. Although FAPEF introduces FFT/IFFT operations at the P4 and P5 stages, these transforms are applied only to lower-resolution semantic feature maps rather than to the highest-resolution P3 layer, which limits the additional computational and memory-transfer cost of the frequency-domain branch.

Table 8. Inference efficiency of the eight ablation models on the NVIDIA GeForce RTX 4070 Laptop GPU.

Model	FAPEF	BiFPN	TCG	Params (M)	GFLOPs	Pre (ms)	Inf (ms)	Post (ms)	Total (ms)	FPS
1				5.0	11.5	1.2	10.6	1.6	13.4	74.6
2	✓			4.7	10.6	1.0	9.7	1.7	12.4	80.6
3		✓		4.3	11.5	0.9	10.2	1.7	12.8	78.1
4			✓	4.2	10.4	0.8	9.2	1.4	11.4	87.7
5	✓	✓		4.2	11.3	0.9	11.3	1.3	13.5	74.1
6	✓		✓	4.1	10.2	0.9	9.6	1.3	11.8	84.8
7		✓	✓	3.7	10.4	1.0	9.9	1.1	12.0	83.3
8	✓	✓	✓	3.6	10.2	0.9	9.2	1.2	11.3	88.5

Note: All models were evaluated with RGB + depth inputs, batch size = 1, input size 640×640 pixels, and FP32 precision. ✓ indicates that the corresponding module is included in the model; Total latency = Pre + Inf + Post; FPS = $1000/\text{Total latency}$. Mean values were obtained over 1000 runs after 50 warmup iterations.

As shown in Table 8, the complete OAFBC-Net, Model 8, achieves the highest inference efficiency among all ablation configurations on the RTX 4070 Laptop GPU. Its total latency is 11.3 ms per image, corresponding to 88.5 FPS. Model 8 also uses the fewest parameters, 3.6 M, and the lowest computational cost, 10.2 GFLOPs, among the eight variants. By comparison, the baseline Model 1 requires 13.4 ms per image and reaches 74.6 FPS, while using 5.0 M parameters and 11.5 GFLOPs. Relative to Model 1, the full model reduces total latency by 15.7% and increases throughput by 18.6%, while simultaneously improving detection accuracy across all occlusion levels reported in Table 5. These results indicate that the proposed modules do not prevent real-time inference on the tested desktop GPU. Instead, the final architecture achieves a better balance among detection accuracy, model size, and runtime performance than the baseline dual-modal detector.

The results also directly address the concern that FFT/IFFT operations in FAPEF may become a computational bottleneck. When FAPEF is introduced alone, Model 2 reduces total latency from 13.4 ms for Model 1 to 12.4 ms, and increases throughput from 74.6 FPS to 80.6 FPS, despite the addition of frequency-domain fusion. More importantly, Model 8 and Model 7 share BiFPN and the TCG Block and differ only in the presence of FAPEF. Under the tested setting, the full model with FAPEF requires 11.3 ms per image and achieves 88.5 FPS; whereas, the variant without FAPEF requires 12.0 ms per image and achieves 83.3 FPS. Therefore, the FFT/IFFT operations in FAPEF do not dominate end-to-end latency and do not create a measurable runtime bottleneck relative to the closest ablated counterpart. This finding is consistent with the efficiency-oriented placement of FAPEF at the lower-resolution P4 and P5 stages described in Section 2.2.

The ablation results further reveal module-specific runtime behaviors. When the TCG Block is introduced alone, Model 4 achieves a total latency of 11.4 ms per image and a throughput of 87.7 FPS, which is the highest among the single-module variants. When FAPEF is combined with the TCG Block, Model 6 maintains a competitive total latency of 11.8 ms per image and a throughput of 84.8 FPS. By contrast, BiFPN increases inference cost more noticeably when combined with FAPEF. Model 5, which integrates FAPEF and BiFPN without the TCG Block, requires 13.5 ms per image and reaches 74.1 FPS, making it the slowest configuration in Table 8. However, when BiFPN is integrated together with FAPEF and the TCG Block in the complete OAFBC-Net, Model 8 reduces total latency to 11.3 ms per image and achieves 88.5 FPS. This suggests that the three modules are jointly optimized in OAFBC-Net and that runtime performance should be evaluated at the complete-system level rather than from an isolated FAPEF implementation alone.

In summary, the inference efficiency experiments support the operational viability of OAFBC-Net on the current RTX 4070 Laptop GPU platform. The complete model achieves 88.5 FPS with 3.6 M parameters and 10.2 GFLOPs, and the measured latencies do not indicate that FAPEF-related FFT/IFFT operations become the dominant runtime bottleneck in the tested dual-branch topology. Nevertheless, these conclusions are limited to the desktop/laptop GPU used in this study. Because edge devices such as NVIDIA Jetson platforms were not available during this revision, future work will extend the same inference protocol to embedded hardware and further evaluate TensorRT or quantization-based deployment for orchard edge perception.

3.8. Visualization Experiments

To further examine the detection behavior of OAFBC-Net, visual comparison experiments are conducted under different input modalities, including RGB, depth, and RGB + depth. The results are shown in Figure 11. YOLOv5, YOLOv8, YOLOv10, YOLOv11, and OAFBC-Net are selected for qualitative comparison.

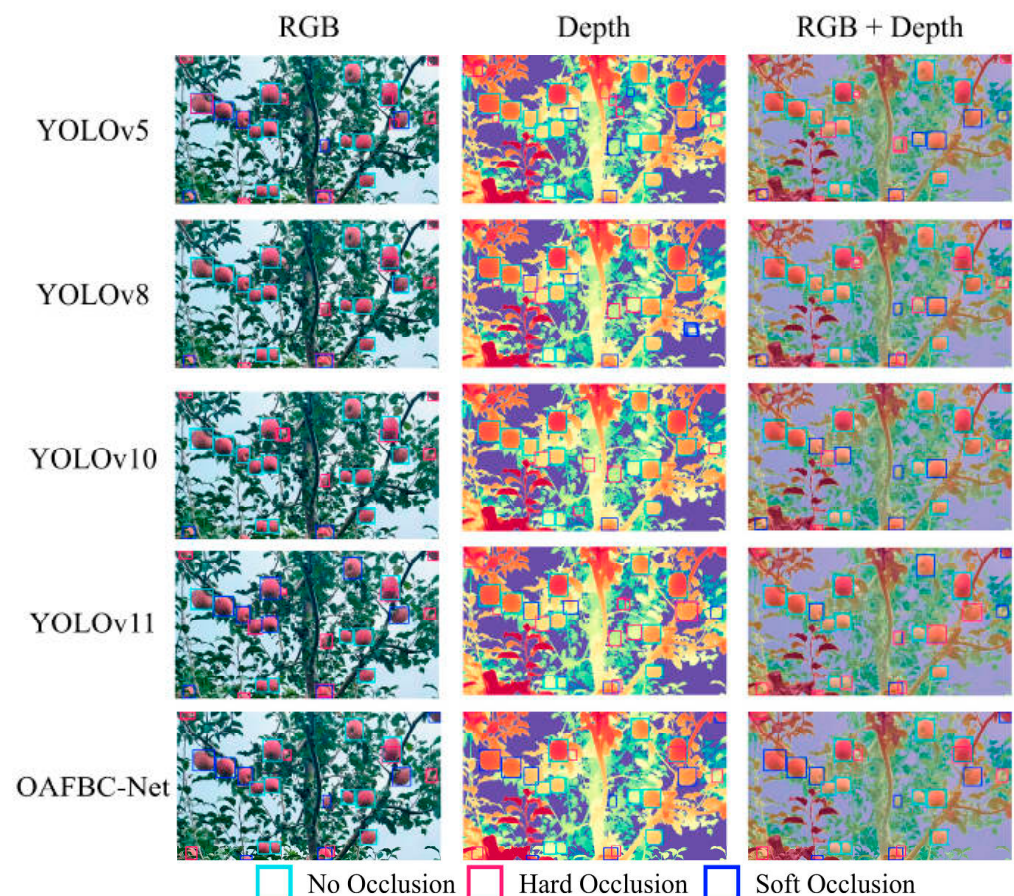


Figure 11. Comparison of detection results produced by YOLOv5, YOLOv8, YOLOv10, YOLOv11, and OAFBC-Net on the Apple dataset using RGB-only, depth-only, and multimodal fusion inputs.

As shown in Figure 11, for non-occluded targets, each detection model generally performs well because the fruit contours are complete and color and texture features are clear. With RGB-only input, the models identify most non-occluded fruits because RGB images contain rich color, texture, and edge information. However, in complex backgrounds or densely distributed fruit regions, some single-modal models still show inaccurate localization or insufficient separation of adjacent targets. By fusing RGB and depth information, OAFBC-Net uses additional spatial-structure cues and improves localization stability for non-occluded targets.

For soft-occluded targets, the fruits are partially occluded by leaves or branches but still retain color, texture, and shape cues. With RGB-only input, the models can detect some soft-occluded fruits, but missed detections and localization deviations occur when leaf colors are similar to the fruit background or when fruit boundaries are partly occluded. With depth-only input, discriminative ability decreases because fine-grained texture and color information are absent. RGB + depth input combines appearance and spatial-structure information, improving the representation of soft-occluded targets. The visual results show that OAFBC-Net reduces some missed detections and improves the match between detection boxes and target regions.

For hard-occluded targets, the visible fruit region is small, the boundary is blurred, and the feature information is incomplete. Detection is therefore more difficult than for non-occluded and soft-occluded targets. Single-modal methods are more vulnerable to leaf occlusion, branch interference, and complex backgrounds, leading to missed detections, false detections, or inaccurate localization. This problem is especially clear with depth-only input. If the depth difference between fruits and surrounding branches or leaves is small,

the model cannot obtain sufficient discriminative features. By fusing RGB texture cues with depth-based spatial-structure cues, OAFBC-Net alleviates this problem to some extent and detects more valid targets in hard-occluded regions.

Overall, RGB + depth multimodal input performs better than a single RGB or depth modality. RGB images provide appearance information, including color, texture, and edges, while depth images provide spatial relationships between targets and backgrounds. Their fusion enhances the model representation for fruits under different occlusion levels, especially in Soft Occlusion and Hard Occlusion scenes where background interference and partial visibility are major challenges.

Among all compared methods, OAFBC-Net shows better visual detection results for soft-occluded, hard-occluded, and non-occluded targets. Compared with YOLOv5, YOLOv8, YOLOv10, and YOLOv11, it produces more complete detections, more accurate bounding boxes, and fewer missed and false detections in the displayed examples. This supports the quantitative results and suggests that multimodal complementary information improves adaptability in complex orchard scenes.

The visualization results further support the effectiveness of OAFBC-Net. For non-occluded targets, the model achieves accurate localization. For soft-occluded targets, it improves detection stability under partial occlusion. For hard-occluded targets, it still detects more valid targets when visible information is limited. A small number of missed and false detections may still occur under extreme occlusion or highly similar backgrounds. Overall, the visualization results are consistent with the quantitative results in Table 4.

To examine the role of FAPEF in multimodal feature fusion, this study visualizes feature responses. As shown in Figure 12, the depth image provides spatial structure differences between fruits and branches or leaves, but its independent representation still suffers from blurred boundaries and background interference. Simple feature concatenation introduces depth information but does not sufficiently filter effective features, resulting in relatively scattered feature responses and limited distinction between target regions and complex backgrounds. In contrast, FAPEF highlights apple regions more clearly and suppresses background noise, such as branches, leaves, and shadows. It also produces clearer responses to fruit edges and local contour details. These results suggest that FAPEF adaptively uses complementary information from RGB and depth features rather than simply stacking them.

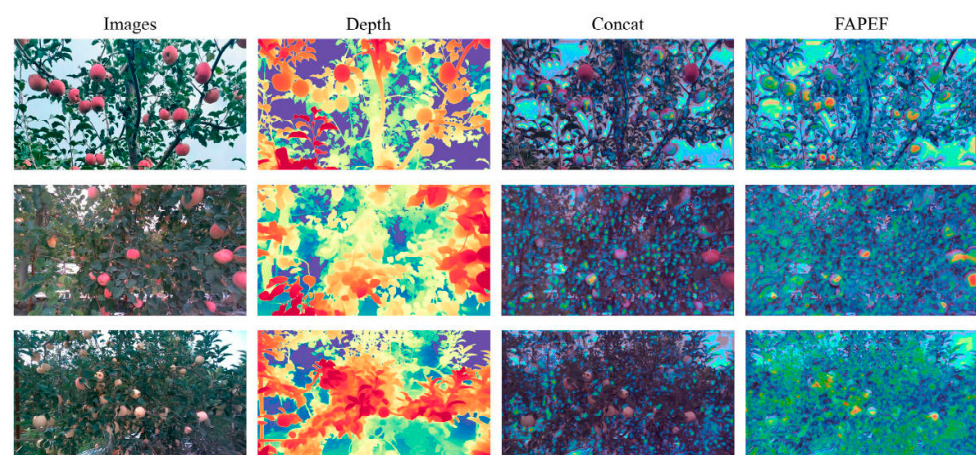


Figure 12. Effect of FAPEF on feature extraction. Brighter colors indicate higher model attention to the corresponding regions.

Figure 13 compares the intermediate features of Base and TCG under the same dual-modal input, including P4 fused features, P5 fused features, and final P5 detection features. TCG produces stronger and more concentrated spatial responses at all three depths, which is consistent with the design goal of the triple TCG Block.

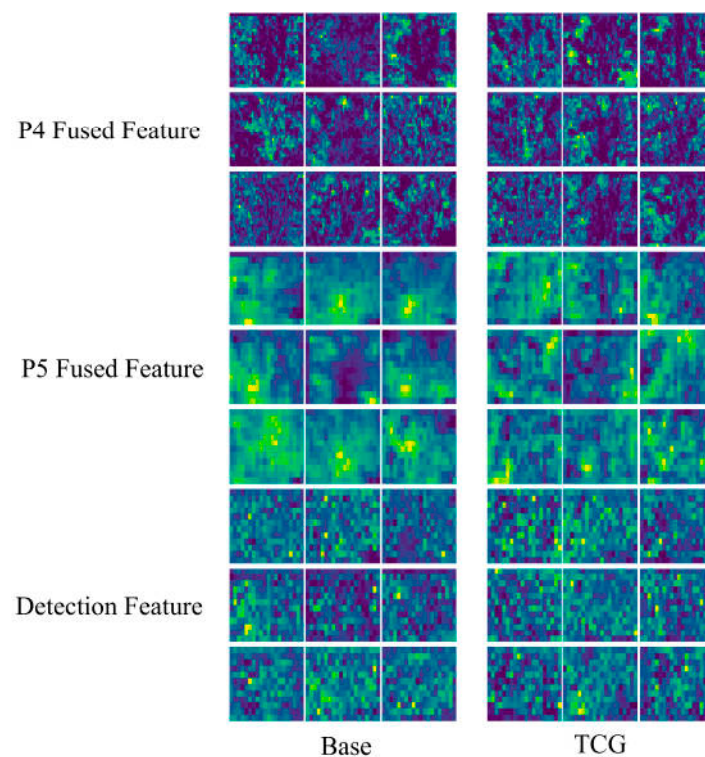


Figure 13. Comparison of intermediate features between Base and TCG on representative samples.

At the P4 fused feature stage, the strong-response channels of the baseline model are relatively sparse, and high-activation regions are mostly scattered. In contrast, TCG presents fuller bright regions with clearer boundaries at the same layer. This indicates that, after RGB and depth information is fused at the P4 scale, features refined by TCG have higher spatial selectivity. At the P5 fused feature stage, the difference becomes more obvious. Base still mainly presents diffuse and block-like high responses; whereas, TCG shows a denser and more structured activation pattern. In the detection features, the strong-response channels of Base are weaker and sparser, while TCG maintains rich and concentrated high-activation regions. This suggests that pre-detection semantic features are better preserved for subsequent bounding box regression and classification.

Each channel is displayed using independent min-max normalization. From the response patterns, TCG maintains more structured feature encoding from the fusion layers to the detection layer. This qualitatively agrees with the improvement observed for the Hard Occlusion category on the validation set. In summary, this visualization suggests that TCG forms stronger and more focused multimodal feature expressions during middle fusion and pre-detection stages.

To further evaluate classification and detection ability under different input modalities and occlusion levels, normalized confusion matrices are plotted in Figure 14. The diagonal values represent the proportion of correctly classified samples in each category; higher values indicate better classification performance. The off-diagonal values represent misclassification proportions among categories and help analyze category confusion under similar occlusion types or complex background conditions. “--” denotes that the corresponding value is not applicable and is excluded from the calculation.

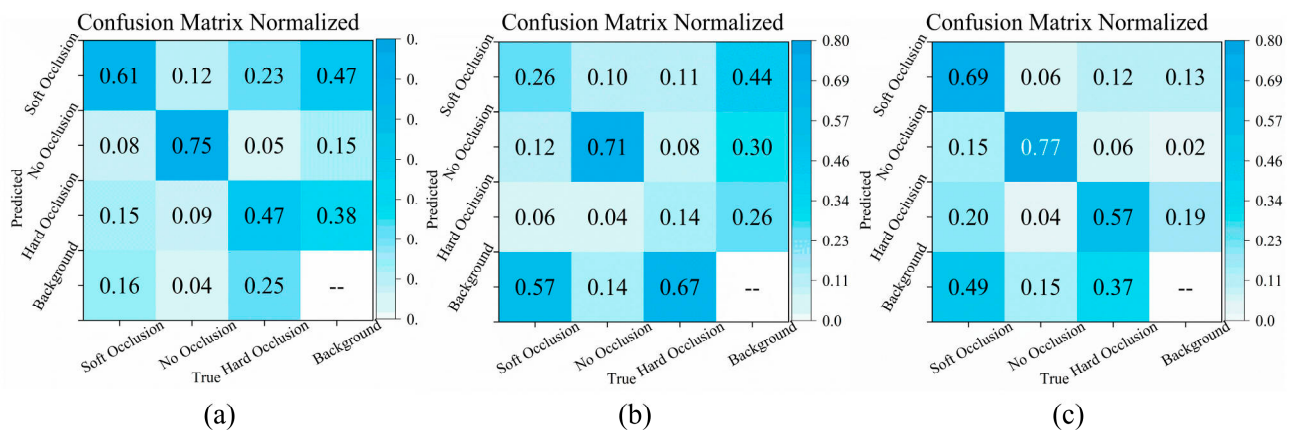


Figure 14. Normalized confusion matrices: (a) YOLOv11n with visible images; (b) YOLOv11n with depth images; (c) OAFBC-Net.

As shown in Figure 14, the visible-light single-modal YOLOv11n has good recognition ability for the No Occlusion category, with a correct classification ratio of 0.75. However, the correct classification ratios for Soft Occlusion and Hard Occlusion are 0.61 and 0.47, respectively, indicating that RGB-only input remains sensitive to leaf occlusion, fruit overlap, and background similarity. The depth-only YOLOv11n obtains correct classification ratios of 0.26, 0.71, and 0.14 for the three apple categories, respectively, showing weaker overall performance. This suggests that depth noise, holes, and boundary discontinuity make it difficult to distinguish occlusion levels when depth images are used alone. In contrast, OAFBC-Net achieves correct classification ratios of 0.69, 0.77, and 0.57 for Soft Occlusion, No Occlusion, and Hard Occlusion, respectively. For Soft Occlusion and Hard Occlusion, OAFBC-Net improves over the visible-light model by 0.08 and 0.10, respectively, indicating that RGB-depth fusion improves occlusion-level discrimination.

Overall, the confusion matrices in Figure 14 show that OAFBC-Net achieves higher classification accuracy and lower category confusion for non-occluded, soft-occluded, and hard-occluded apples. The advantage is most evident for Hard Occlusion. These results suggest that the multimodal fusion strategy and feature enhancement modules improve robustness in complex occlusion scenes and may provide useful visual perception results for target localization, occlusion assessment, and grasping decisions in orchard picking robots.

3.9. Analysis of Depth Sensor Effects Under Direct Natural Illumination

The Intel RealSense D455 utilized in this study is an active stereo depth camera that estimates depth through infrared pattern projection and stereo matching. This category of sensors is highly sensitive to two factors in outdoor orchard environments, which are sunlight degradation, where direct sunlight saturates the infrared receiver and invalidates projected patterns to reduce the valid pixel ratio in depth maps, and organic leaf structural voids, where thin organic structures such as leaves often manifest as fragmented, missing, and discontinuous boundaries rather than intact continuous surfaces due to surface reflection, translucency, and complex microstructures. Since OAFBC-Net relies on the cross-modal fusion of RGB and depth data, such hardware-level depth invalidation can directly affect the detection performance of the Hard Occlusion category.

On the full test set, OAFBC-Net achieves AP50 scores of 87.3%, 72.1%, and 57.9% for the No Occlusion, Soft Occlusion, and Hard Occlusion categories, respectively, as presented in Table 4. Evidently, the performance for the Hard Occlusion category is substantially lower than that for the other two categories, revealing a significant performance gap. It is necessary to further differentiate the extent to which this gap stems from insufficient model learning of severely occluded targets versus depth observation failure of the D455 under

direct light and organic leaf structural voids. To address this, this section selects 300 Direct Sunlight images and 300 Non-direct or Diffuse images from the independent test set of the complete dataset based on illumination conditions to construct an illumination-stratified subset. Statistical analysis is then conducted at both the depth quality and detection performance levels to quantify the contribution of sensor limitations to the Hard Occlusion performance gap.

Depth quality metrics are computed on the raw depth maps to reflect the hardware observation failure of the RealSense D455. The image-level Valid Depth Ratio, denoted as VDR, is defined as the proportion of valid pixels in the raw depth map relative to the full image. The bounding-box Invalid Depth Ratio, denoted as IDR, is defined as the proportion of invalid pixels within each ground-truth bounding box relative to the total box pixels, with mean values calculated separately for the No Occlusion, Soft Occlusion, and Hard Occlusion categories. Invalid pixels are identified as those with a depth value of 0 in the depth map.

Table 9 presents the depth quality statistics for the two groups. The mean VDR of the Direct Sunlight group is 47.86%, which is lower than the 52.12% of the Non-direct or Diffuse group, indicating a decline in full-image depth validity under direct light conditions. At the target level, the mean IDR within Hard Occlusion bounding boxes increases from 39.41% under Non-direct or Diffuse conditions to 54.73% under Direct Sunlight, representing an increase of 15.32 percentage points. The increases in IDR within Soft Occlusion and No Occlusion bounding boxes are 6.78 and 4.86 percentage points, respectively. Both groups satisfy the condition $IDR_{NO} < IDR_{SO} < IDR_{HO}$, which demonstrates that depth invalidation is most severe in Hard Occlusion target regions and that this effect is further amplified under Direct Sunlight conditions. This implies that the sensitivity of the D455 to sunlight degradation and organic leaf structural voids is most pronounced in the Hard Occlusion category.

Table 9. Depth quality statistics under different illumination conditions.

Illumination Group	Test Images	Mean VDR (%)	IDR NO (%)	IDR SO (%)	IDR HO (%)
Direct Sunlight (DS)	300	47.86	36.51	45.77	54.73
Non-direct/Diffuse (ND)	300	52.12	31.65	38.99	39.41

Note: VDR denotes the image-level Valid Depth Ratio. IDR denotes the Invalid Depth Ratio inside ground-truth bounding boxes. All statistics in Table 9 were computed on raw depth maps before hole filling. Detection results in Table 10 used the same preprocessed depth maps as in the main experiments.

Table 10. Occlusion-aware detection performance under different illumination conditions.

Illumination Group	Model	AP50(%)			HO–NO Gap (pp)
		SO	NO	HO	
Direct Sunlight (DS)	RGB-only YOLOv11	64.1	81.8	46.2	35.6
Non-direct/Diffuse (ND)	RGB-only YOLOv11	66.5	83.5	48.1	35.4
Direct Sunlight (DS)	OAFBC-Net	70.3	86.2	54.5	31.7
Non-direct/Diffuse (ND)	OAFBC-Net	74.6	88.7	61.8	26.9

Note: HO–NO gap = AP50_NO – AP50_HO. All results were obtained on the same 600-image independent test set used in Tables 4 and 5, partitioned into 300 DS and 300 ND images.

The AP50 scores of the three occlusion categories for RGB-only YOLOv11 and OAFBC-Net are further evaluated on the same Direct Sunlight and Non-direct or Diffuse subsets as presented in Table 10. Since the RGB-only model operates independently of depth information, its Hard Occlusion AP50 decreases marginally from 48.1% under Non-direct or Diffuse conditions to 46.2% under Direct Sunlight conditions. This minimal drop of 1.9 percentage points indicates an insensitivity to illumination variations. Conversely, OAFBC-Net relies

on RGB-depth fusion. Its Hard Occlusion AP50 experiences a more substantial decline of 7.3 percentage points from 61.8% under Non-direct or Diffuse lighting to 54.5% under Direct Sunlight. The corresponding performance drops for the Soft Occlusion and No Occlusion categories are 4.3 and 2.5 percentage points, respectively. The observation that the Hard Occlusion category exhibits the highest sensitivity to illumination changes aligns well with the maximal increase in the Invalid Depth Ratio within Hard Occlusion bounding boxes reported in Table 9.

Analyzing the performance gap between the Hard Occlusion and No Occlusion categories reveals that the gap for the RGB-only model remains relatively constant at approximately 35.6 and 35.4 percentage points under Direct Sunlight and Non-direct or Diffuse conditions, respectively. For OAFBC-Net, this gap measures 31.7 percentage points under Direct Sunlight but narrows to 26.9 percentage points under Non-direct or Diffuse conditions. This reduction demonstrates that multi-modal fusion effectively bridges the performance divide between heavily occluded and unoccluded targets when depth quality is favorable. In addition, compared to the RGB-only baseline, OAFBC-Net achieves a Hard Occlusion performance gain of 8.3 percentage points under Direct Sunlight and a larger gain of 13.7 percentage points under Non-direct or Diffuse conditions. This indicates that the model derives greater benefits from cross-modal fusion as depth quality improves.

By synthesizing the data from Tables 9 and 10, the underperformance of the Hard Occlusion category relative to the No Occlusion and Soft Occlusion categories can be attributed to three interconnected factors. The first factor is the intrinsic difficulty of the task. Even under favorable Non-direct or Diffuse conditions, the Hard Occlusion AP50 of OAFBC-Net remains at 61.8%, which is considerably lower than the 88.7% achieved for the No Occlusion category. This suggests that severe occlusion inherently constitutes a detection bottleneck that is not exclusively caused by sensor or model limitations. The second factor involves the hardware constraints of the RealSense D455 camera. Under Direct Sunlight, the Valid Depth Ratio decreases and the Invalid Depth Ratio within Hard Occlusion bounding boxes increases significantly, which corresponds to a 7.3 percentage point drop in the Hard Occlusion AP50 for OAFBC-Net. In contrast, the RGB-only model remains largely unaffected. This contrast proves that a substantial portion of the Hard Occlusion performance gap is related to depth invalidation caused by direct light and organic leaf structural voids rather than purely inadequate network design. The third factor is the fusion capability of the model. OAFBC-Net consistently outperforms the RGB-only model in Hard Occlusion detection under both illumination conditions, with a more pronounced fusion gain under Non-direct or Diffuse lighting. This confirms that modules such as FAPEF can successfully extract useful geometric information when depth data is partially valid. However, the model cannot completely compensate for hardware observation deficits when the depth information within a bounding box is extensively invalid. Consequently, the overall Hard Occlusion AP50 of 57.9% on the complete test set, which is significantly lower than that of the other two categories, should be interpreted as the combined result of intrinsic occlusion difficulty, outdoor depth observation failures of the D455 camera, and the upper bounds of network fusion capabilities. It should not be attributed solely to insufficient model capacity. The statistical results obtained from the illumination-stratified subsets provide a firm quantitative foundation for this comprehensive attribution.

4. Discussion

OAFBC-Net is developed for occlusion-aware apple detection in semi-structured orchard canopies, where RGB-only methods often miss or mislocalize partially hidden fruit. On the single-site RGB-depth dataset, integrating FAPEF, BiFPN, and TCG improves

AP50 for hard-occluded apples compared with concatenation-based RGB-depth baselines, suggesting that frequency-domain cross-modal fusion and context modeling help disambiguate canopy clutter. These conclusions are limited to the reported orchard location, imaging setup, and annotation protocol. Cross-orchard, cross-cultivar, and field-robot trials are still required before operational readiness for autonomous harvesting can be claimed.

Compared with other object detection algorithms, OAFBC-Net achieves better overall performance in the occluded apple detection task. In the comparative experiments among RGB-only, depth-only, and RGB + depth detection methods, OAFBC-Net achieves AP50 values of 72.1%, 87.3%, and 57.9% for soft-occluded, non-occluded, and hard-occluded apples, respectively. In particular, for hard-occluded apple detection, OAFBC-Net shows a clear advantage over the self-implemented multimodal YOLO baselines. These results indicate that frequency-domain enhancement, context modeling, and multi-scale fusion improve the ability of the model to recognize occluded targets in this dataset. Meanwhile, OAFBC-Net has only 3.6 M parameters, indicating a lightweight structure with competitive accuracy.

Model stability is further examined through multi-seed experiments and five-fold cross-validation. The multi-seed experiments show that OAFBC-Net maintains stable detection performance under different random initialization conditions, indicating that its performance gain is not tied to a single random seed. The five-fold cross-validation results show consistent performance under different training and validation splits. These results provide additional evidence for reproducibility and robustness to data partitioning, while remaining limited to the collected dataset and fixed test set. They also indicate that FAPEF, BiFPN, and TCG improve detection accuracy without obvious increases in training-result instability.

Although the proposed method performs well on the apple RGB-depth multimodal occlusion detection dataset, it still has several limitations. First, the data are mainly collected from specific orchard environments. Although the dataset contains different illumination conditions, occlusion types, and complex backgrounds, the generalization ability of the model across different regions, apple varieties, tree structures, and growth stages still requires validation. Second, depth images are affected by sensor noise, holes, strong light interference, and acquisition-distance changes. When the spatial depth difference between fruits and branches or leaves is small, the model may still produce missed or false detections. Section 3.9 further shows that direct sunlight increases Invalid Depth Ratios in Hard Occlusion bounding boxes and corresponds to a measurable Hard Occlusion AP50 drop of 7.3 percentage points for OAFBC-Net, indicating that part of the HO performance gap is associated with RealSense D455 depth invalidation under outdoor active stereo sensing rather than with network design alone. In addition, although OAFBC-Net is lightweight in parameter size, Section 3.7 shows that the complete model achieves a total latency of 11.3 ms per image and 88.5 FPS on the NVIDIA GeForce RTX 4070 Laptop GPU under batch size = 1 and input size 640×640 pixels. These results support real-time inference on the tested desktop GPU platform. However, deployment efficiency and stability on low-power embedded platforms, mobile terminals, and real picking robot systems, including NVIDIA Jetson devices, still require further validation. In more complex natural environments, such as rain, fog, strong backlight, nighttime operation, severe branch and leaf occlusion, and dense fruit clusters, model robustness still needs improvement.

Future work will explore pruning, knowledge distillation, model quantization, and inference acceleration to reduce the computational complexity and storage cost of OAFBC-Net and improve its real-time detection ability on embedded devices and orchard picking robot platforms. To improve generalization in complex orchard environments, the RGB-depth apple occlusion detection dataset will be expanded with samples from different orchards, varieties, weather conditions, illumination conditions, and growth stages. Future

research will also consider additional modalities, such as near-infrared, thermal infrared, or multispectral information, fused with RGB and depth data to improve target detection under low illumination, severe occlusion, and complex backgrounds. Finally, OAFBC-Net will be integrated with 3D localization, picking-point estimation, robotic-arm path planning, and obstacle-avoidance control to evaluate its practical value in autonomous orchard picking tasks.

5. Conclusions

This paper proposes OAFBC-Net, an RGB-depth multimodal detection network for occluded apple detection and occlusion-level recognition in complex orchard environments. The method aims to improve the detection accuracy and robustness of non-occluded, soft-occluded, and hard-occluded apple targets in natural orchards. To address missed detections, false detections, and unstable localization that commonly occur in RGB-only detection under illumination variation, branch and leaf occlusion, fruit overlap, and background similarity, this paper introduces depth images as a supplement to spatial geometric information and enhances the perception of occluded apples through cross-modal feature fusion.

In the feature fusion stage, this paper designs the FAPEF Fourier amplitude-phase enhancement fusion module. FAPEF maps RGB and depth features into the frequency domain and uses amplitude and phase information to enhance fruit edge, texture, contour, and spatial-structure representations. In the multi-scale feature fusion stage, BiFPN achieves adaptive fusion among features at different scales through learnable weights and strengthens the interaction between shallow detail information and deep semantic information. TCG jointly models local, surrounding, and global contexts to supplement missing semantic cues under occlusion and improve the discrimination of soft-occluded and hard-occluded apples.

The experimental results show that OAFBC-Net achieves strong performance on the apple RGB-depth occlusion detection dataset. Compared with the YOLO series, SSD, RetinaNet, Faster R-CNN, and several multimodal YOLO baselines under the tested settings, OAFBC-Net achieves AP50 values of 72.1%, 87.3%, and 57.9% for soft-occluded, non-occluded, and hard-occluded apples, respectively. Ablation experiments show that FAPEF, BiFPN, and TCG each contribute to model performance. When the three modules are used together, Precision, Recall, mAP50, and mAP50-95 reach 64.8%, 71.7%, 72.4%, and 55.2%, respectively, while the parameter count is reduced to 3.6 M compared with the baseline. The visualization results also show that OAFBC-Net focuses more accurately on apple target regions and reduces some missed and false detections under complex occlusion conditions.

The multi-seed experiments and five-fold cross-validation results show that OAFBC-Net maintains stable performance under different random initialization and data partitioning conditions. This indicates that the performance improvement has good reproducibility within the tested dataset and protocol. The inference efficiency experiments in Section 3.7 further show that the complete model achieves 88.5 FPS with 3.6 M parameters on the RTX 4070 Laptop GPU, indicating that FAPEF-related FFT/IFFT operations do not prevent real-time inference on the tested desktop platform. The illumination-stratified analysis in Section 3.9 further indicates that Hard Occlusion performance degradation is associated with D455 depth invalidation under direct sunlight, in addition to task difficulty and fusion limitations. These results further support the effectiveness of RGB-depth cross-modal fusion and occlusion feature enhancement in complex orchard scenes.

In summary, OAFBC-Net provides an effective RGB-depth fusion method for occluded apple detection in complex orchard environments and offers a technical reference for high-

precision visual perception in orchard picking robots. The method improves occluded-target detection while considering lightweight model design, and the stability, runtime, and illumination-stratified attribution experiments in Sections 3.7 and 3.9 support its application potential on the tested desktop GPU and clarify the respective roles of task difficulty, D455 depth invalidation, and network fusion in Hard Occlusion detection. Future work should extend the same inference protocol to embedded hardware such as NVIDIA Jetson platforms, and combine the method with model compression, multimodal expansion, and robotic closed-loop control to support more stable and efficient automated orchard picking.

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Abbreviations

The following abbreviations are used in this manuscript:

AP50	Average precision at intersection over union (IoU) = 0.50
BiFPN	Bidirectional feature pyramid network
CIoU	Complete intersection over union
DyT	Dynamic Tanh
FAPEF	Fourier amplitude-phase enhancement fusion
FFT	Fast Fourier transform
FPS	Frames per second
GFLOPs	Giga floating-point operations
HO	Hard Occlusion/hard-occluded
IDR	Invalid Depth Ratio
IFFT	Inverse fast Fourier transform
IoU	Intersection over union
mAP50	Mean average precision at IoU = 0.50
mAP50-95	Mean average precision averaged over IoU thresholds from 0.50 to 0.95
NO	No Occlusion/non-occluded
OAFBC-Net	Occlusion-aware RGB-depth fusion network proposed in this study
RGB	Red, green, and blue
RGB-D	Red, green, blue, and depth
RGB-depth	Red, green, blue, and depth
SO	Soft Occlusion/soft-occluded
TCG	Triple context guided
TP	True Positive
FP	False Positive
FN	False Negative
VDR	Valid Depth Ratio

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